



Fundamentals of Optics

Lecturer

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Room 239, South Tower, College of Engineering

Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

CHAPTER 3

THE PROPAGATION OF LIGHT

3.1 Introduction



Reflection (反射)



Transmission (透射)



Refraction (折射)

Scattering

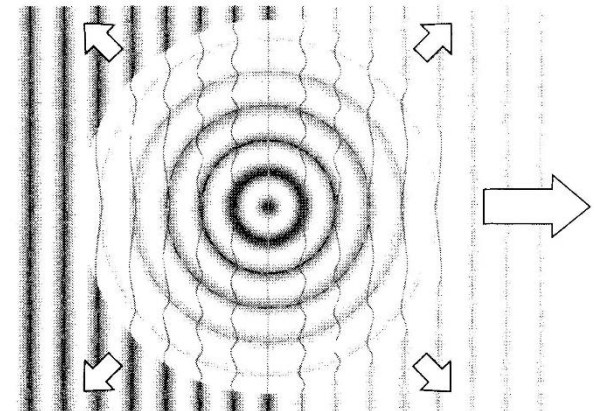
□ **Scattering:** the absorption and prompt re-emission of EM-radiation by electrons associated with atoms and molecules.

The process of transmission, reflection, and refraction are macroscopic manifestations of scattering occurring on a submicroscopic level.

3.2 Rayleigh Scattering

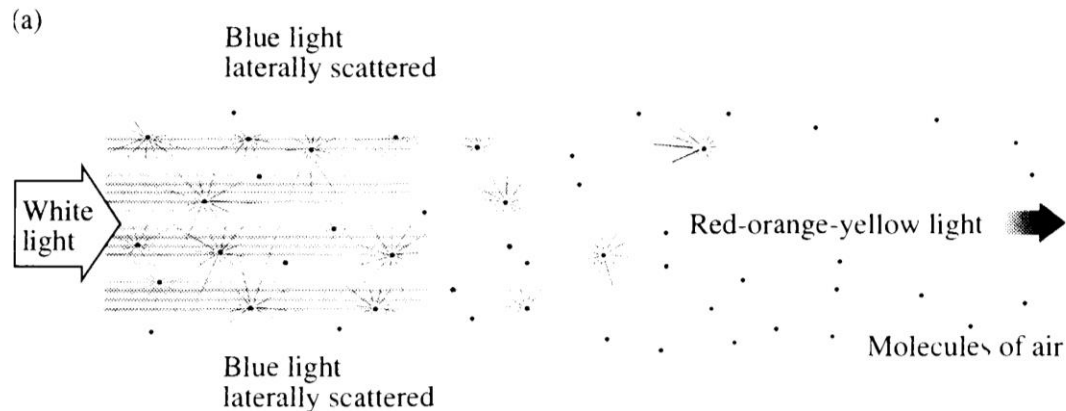
- ❑ Consider a wisp of air (particles smaller than a wavelength)
 - ✓ In the visible, these molecules have no resonances. It means none of them can be raised into an excited state by absorbing a quantum of light, and the gas is therefore transparent.
 - ✓ Instead, each molecule behaves as a little oscillator whose electron cloud can be driven into a ground-state vibration by an incoming photon.
 - ✓ Immediately upon being set vibrating, the molecule initiates the re-emission of light.
 - ✓ A photon is absorbed, and without delay another photon of the same frequency is emitted.

❑ The light is **elastically scattered**



Wave dependence of scattering

All the molecules have electronic resonances in the UV.
The closer the driving frequency is to a resonance, the more vigorously the oscillator responds



Intensity of the light scattered: $I \propto \frac{1}{\lambda^4}$

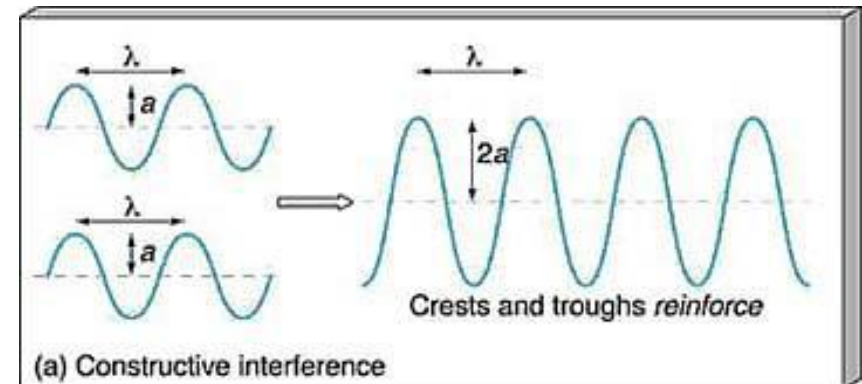
Case: $I_{(\text{blue light } 430\text{nm})} \approx 6 \times I_{(\text{red light } 680\text{nm})}$



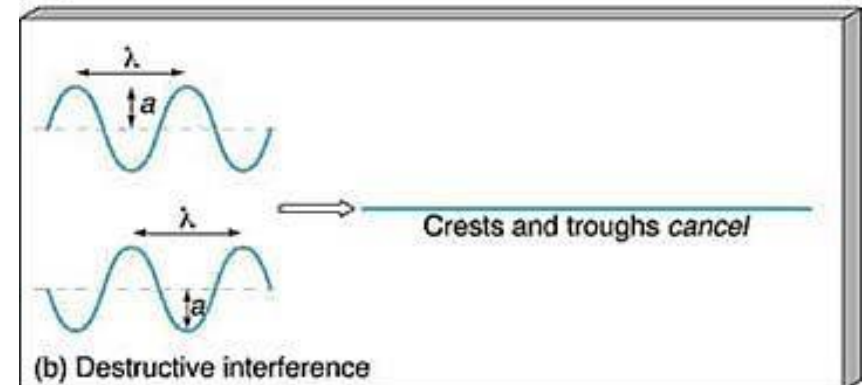
3.2.1 Scattering and interference

❑ Interference: the superposition of two or more waves producing a resultant disturbance that is the sum of the overlapping wave contributions.

❑ Total constructive interference:
in-phase



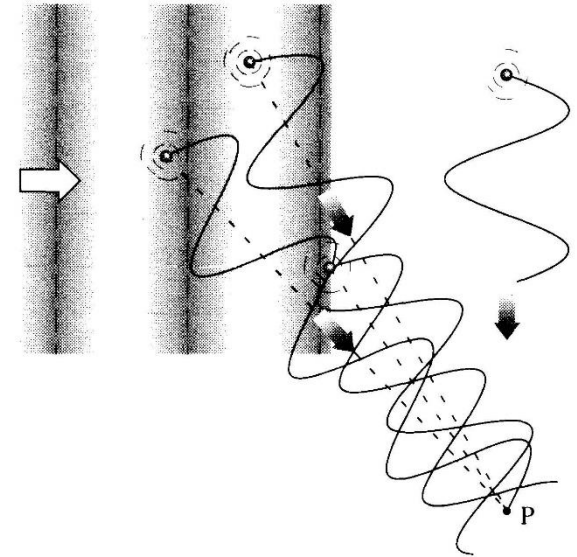
❑ Total destructive interference:
 180° out-of-phase



Tenuous gas: molecules randomly arrayed in space

The wavelets arriving at a lateral point P have a jumble of different phases and tend not to interfere in a sustained constructive fashion.

侧边的。
一团(乱七八糟的)
持续的相长干涉



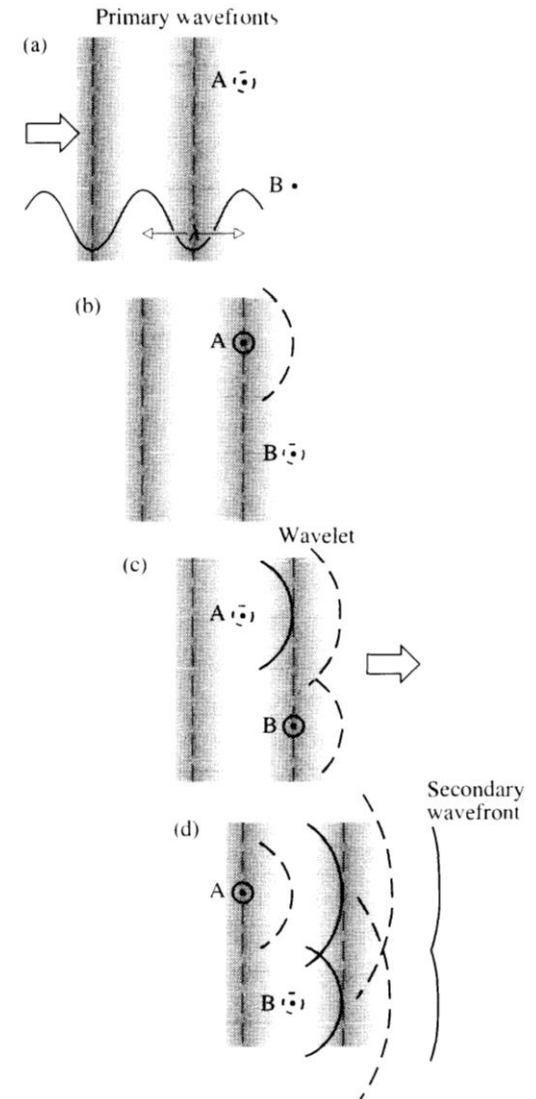
Random, widely spaced scatters driven by an incident primary wave emit wavelets that are essentially independent of one another in all direction except forward. **Laterally scattered light, unimpeded by interference, streams out of the beam.**

随机、广泛间隔的

散射

All the scattered wavelets add **constructively** with each other in the **forward direction**.

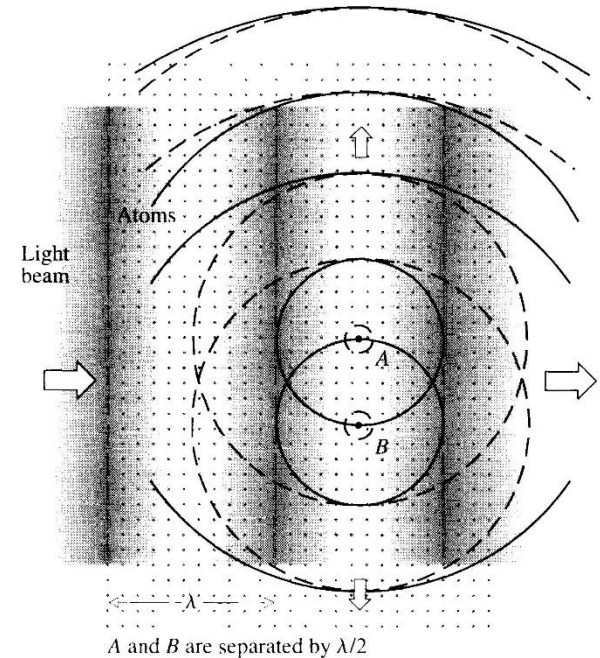
It is also true for dense media



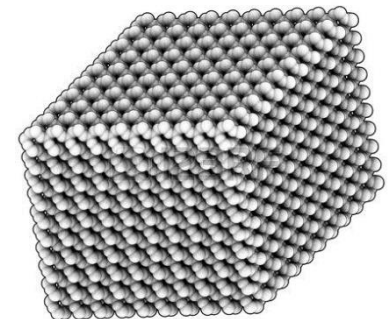
3.2.2 The transmission of light through dense media

Little or no light ends up scattered laterally or backwards in a dense homogeneous medium.

Interference produces a redistribution of energy, out of the regions where it's destructive into the regions where it's constructive.



Example: Good crystals (quartz) scatter even more faintly



3.2.3 Transmission and the index of refraction

- ❑ Secondary wave: the scattered wavelets combined in-phase in the forward direction.
- ❑ Primary wave: the incident electromagnetic wave.
- ❑ Transmitted wave: primary wave + secondary wave

The primary and secondary electromagnetic waves propagate through the interatomic void with the speed c .

But the velocity in medium $v = \frac{c}{n}$

Source: the phase relationship between the secondary and primary waves

Phase shift

Classical forced oscillator:

Opposing restoring force

$$q_e E(t) - m_e \omega_0^2 x - m_e \gamma \frac{dx}{dt} = m_e \frac{d^2 x}{dt^2}$$

Driving force

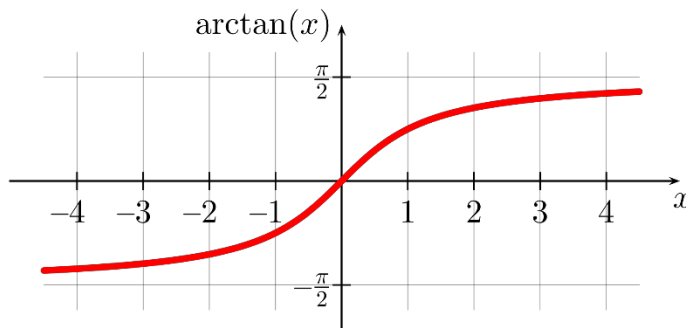
Damping force, energy lost when the forced oscillators re-radiate.

Let $E(t) = E_0 e^{i\omega t}$

The solution: $x = x_0 e^{i(\omega t - \alpha)}$

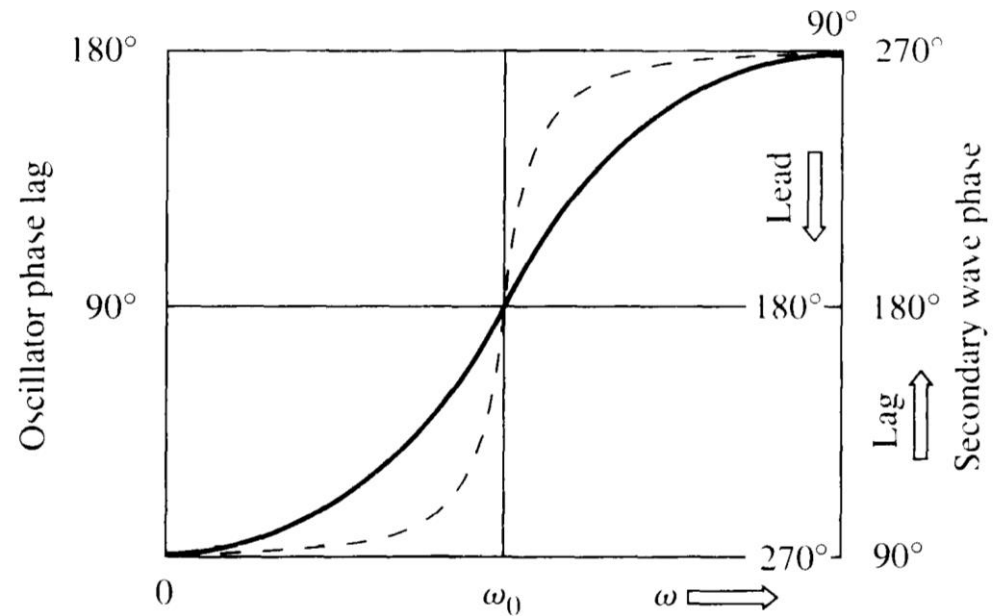
$$x_0 = \frac{q_e E_0}{m_e} \frac{1}{[(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2]^{1/2}}$$

$$\alpha = \tan^{-1} \left[\frac{\gamma \omega}{\omega_0^2 - \omega^2} \right]$$



$$\tan \alpha = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$$

The resultant secondary wave itself lags the oscillators by 90° .
(Homework: Problem 4.5)

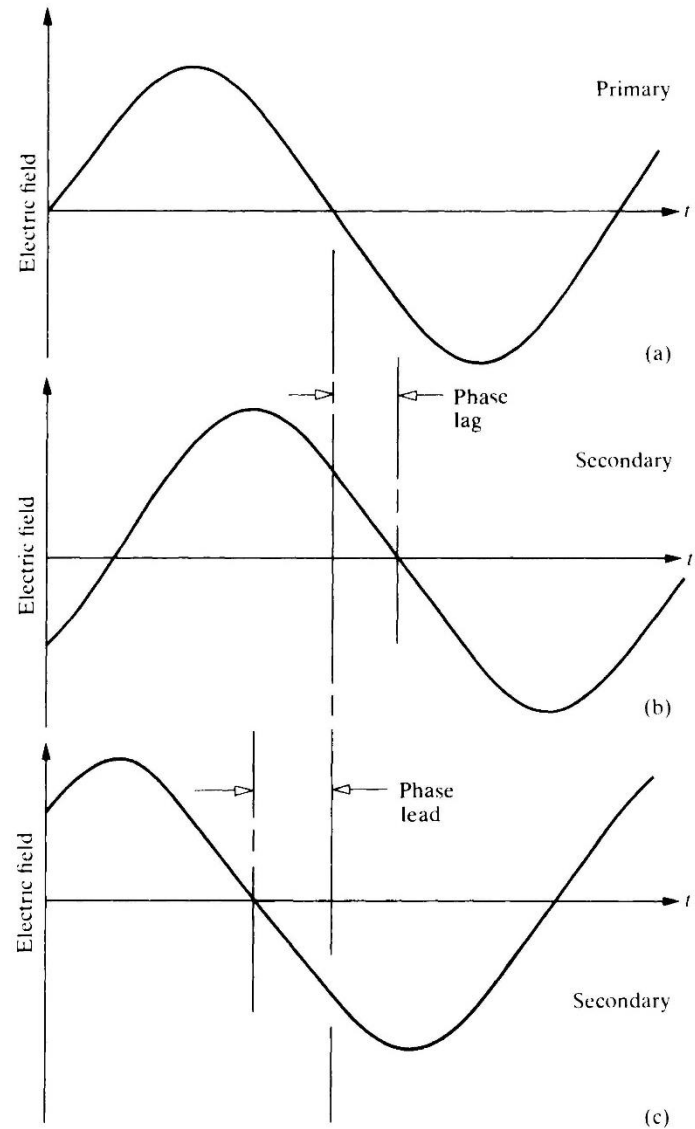


- ❑ At frequency below resonance: the secondary wave lags the primary by some amount between approximately 90° and 180°
- ❑ At frequency above resonance: the secondary wave lags the primary by some amount between approximately 180° and 270°
- ❑ Note: a phase lag of $\delta \geq 180^\circ$ is equivalent to a phase lead of $360^\circ - \delta$

❑ A primary wave

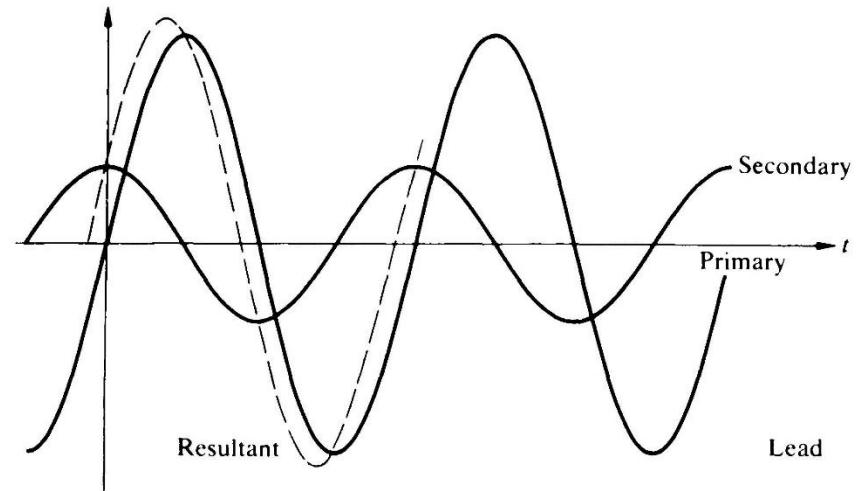
❑ The secondary lags the primary-
it takes longer to reach any
given value.

❑ The secondary wave reaches
any given value before the
primary; that is, it leads.



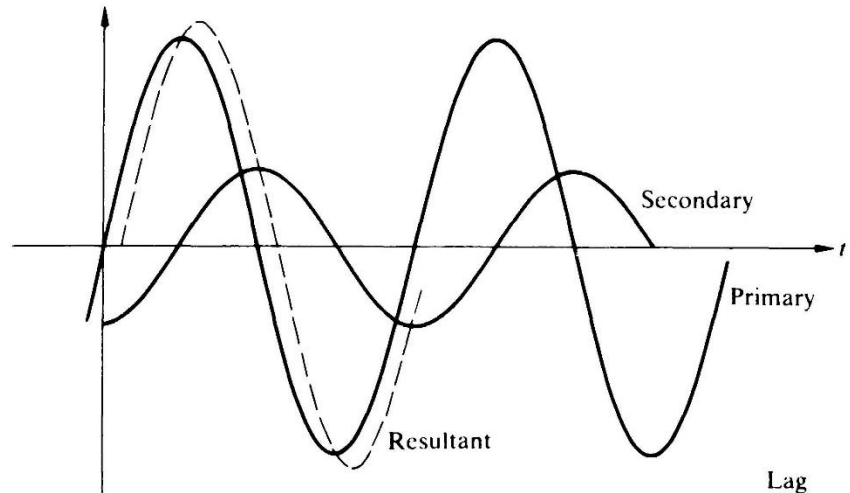
❑ When the secondary wave leads the primary, the resultant transmitted wave leads it by some amount.

- ✓ $\omega > \omega_0$: the transmitted wave leads the free-space wave
- ✓ A phase lead yields an increase in speed, $v > c$ and $n < 1$



❑ When the secondary wave lags the primary, the resultant transmitted wave lags it by some amount.

- ✓ $\omega < \omega_0$: the transmitted wave lags the free-space wave
- ✓ A phase lag corresponds to a reduction in speed, $v < c$ and $n > 1$



Homework

Problem 4.5

4.5 Imagine that we have a nonabsorbing glass plate of index n and thickness Δy , which stands between a source S and an observer P .

- (a) If the unobstructed wave (without the plate present) is $E_u = E_0 \exp i\omega(t - y/c)$, show that with the plate in place the observer sees a wave

$$E_p = E_0 \exp i\omega[t - (n - 1) \Delta y/c - y/c]$$

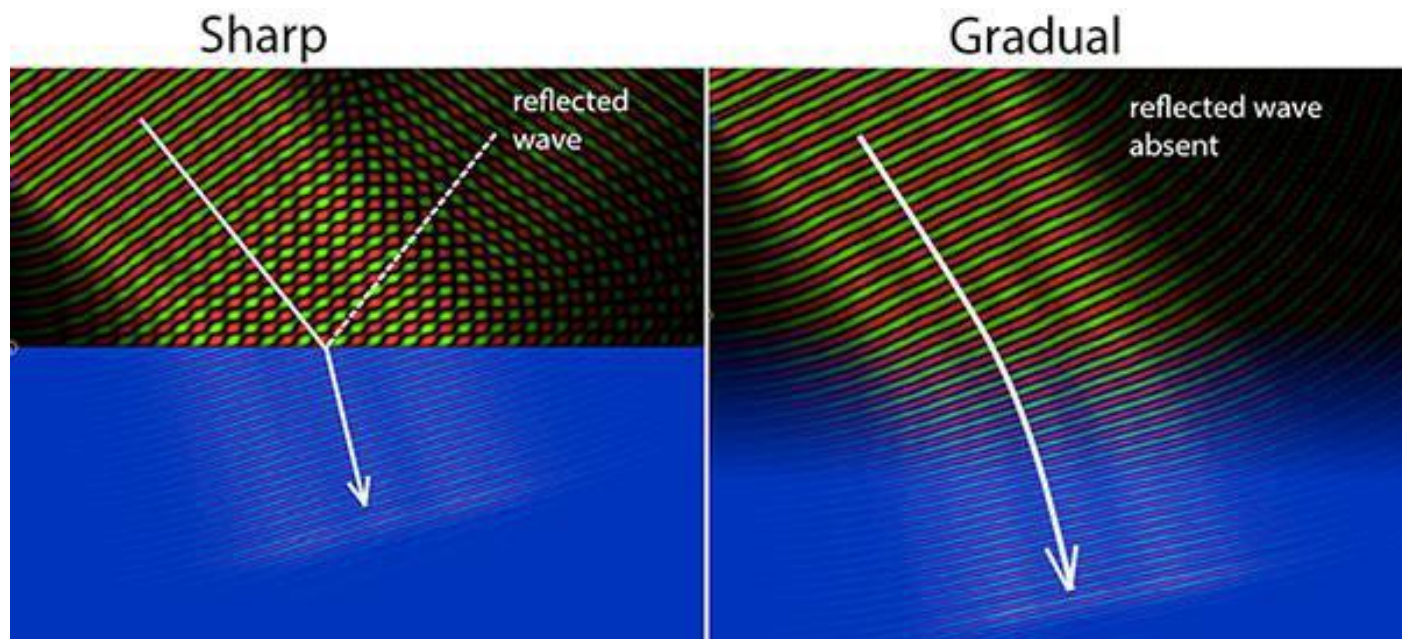
- (b) Show that if either $n \approx 1$ or Δy is very small, then

$$E_p = E_u + \frac{\omega(n - 1) \Delta y}{c} E_u e^{-i\pi/2}$$

The second term on the right may be envisioned as the field arising from the oscillators in the plate.

Reflection

- ❑ When a beam of light strikes an interface, some light is always scattered backward and we call this phenomenon **Reflection**.



- ❖ The transition between two gradual media, there will be very little reflection.

Reflection

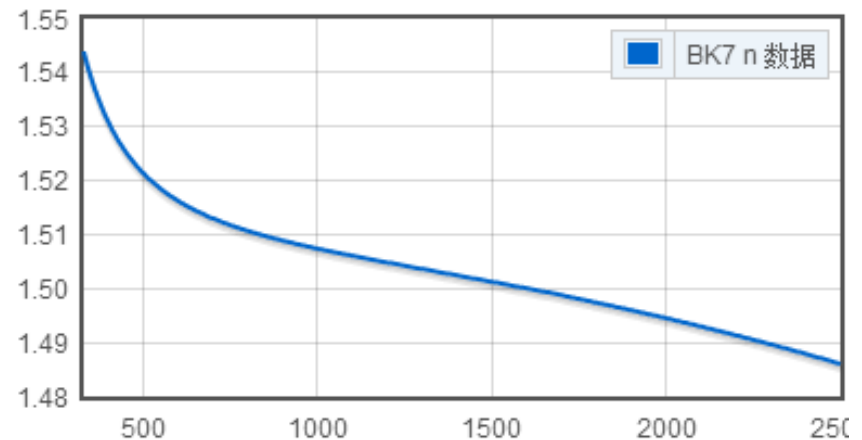
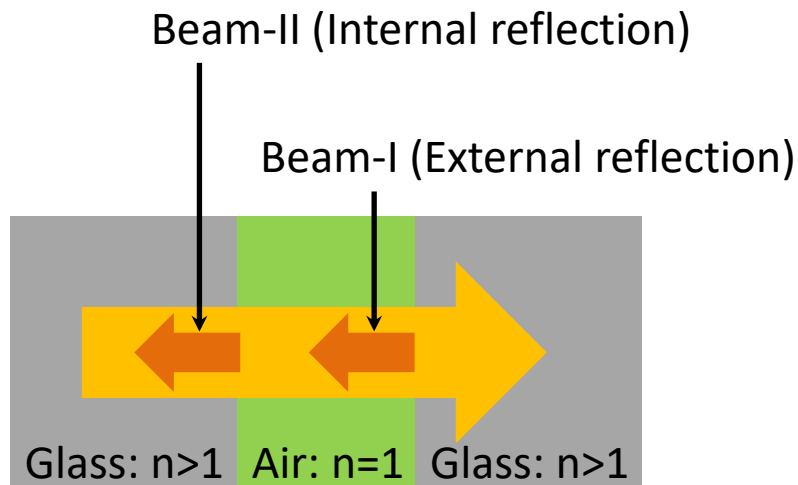
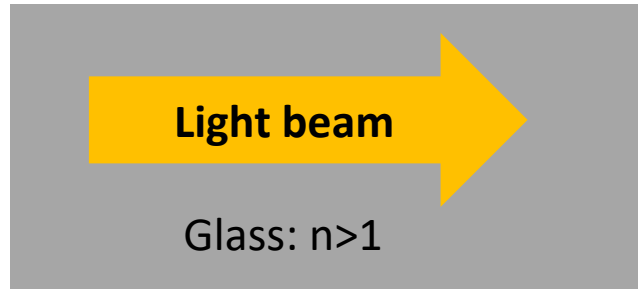


Each molecule scatters light in the backward direction, and in principle, each and every molecule contributes to the reflected wave.

In practice, it is a thin layer ($\approx \lambda/2$ deep) of unpaired atomic oscillators near the surface that is effectively responsible for the reflection.

Example: an air-glass interface, about **4%** of the energy of an incident beam falling perpendicularly in air on glass will be reflected straight back out by this layer of unpaired scatters.

Internal and external reflection



- ❑ **External reflection:** light is traveling from a less to a more optically dense medium ($n_i < n_t$)
- ❑ **Internal reflection:** ($n_i > n_t$)

❖ Note: 180° relative phase shift between internally and externally reflected light

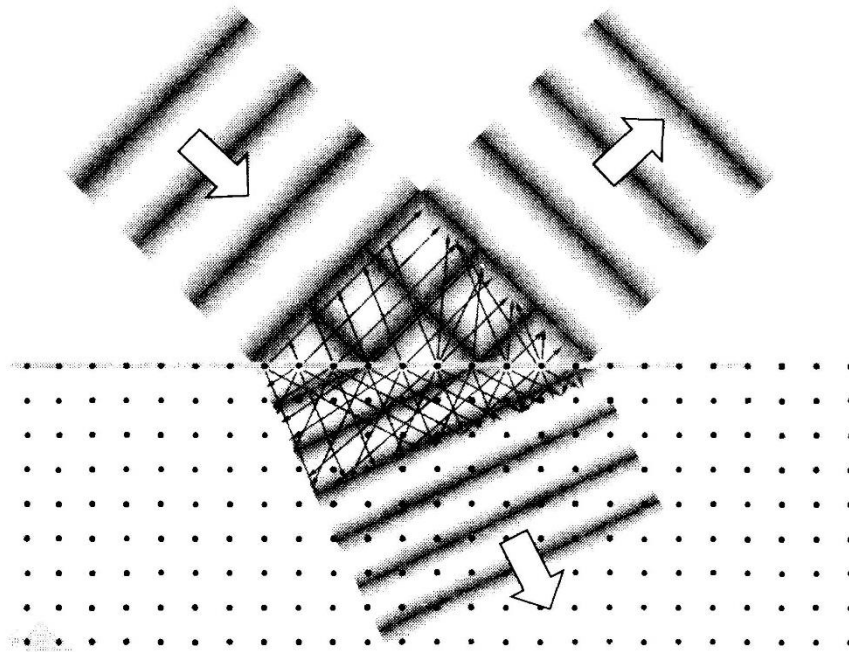
White light is reflected as white

- ❑ The layer of scatterers responsible for the reflection is effectively about $\lambda/2$ thick. Hence, the larger the wavelength, the deeper the region contributing, and the more scatterers there are acting together.
- ❑ Each scatter is less efficient as λ increase ($\propto 1/\lambda^4$)

Two effects tend to balance out.

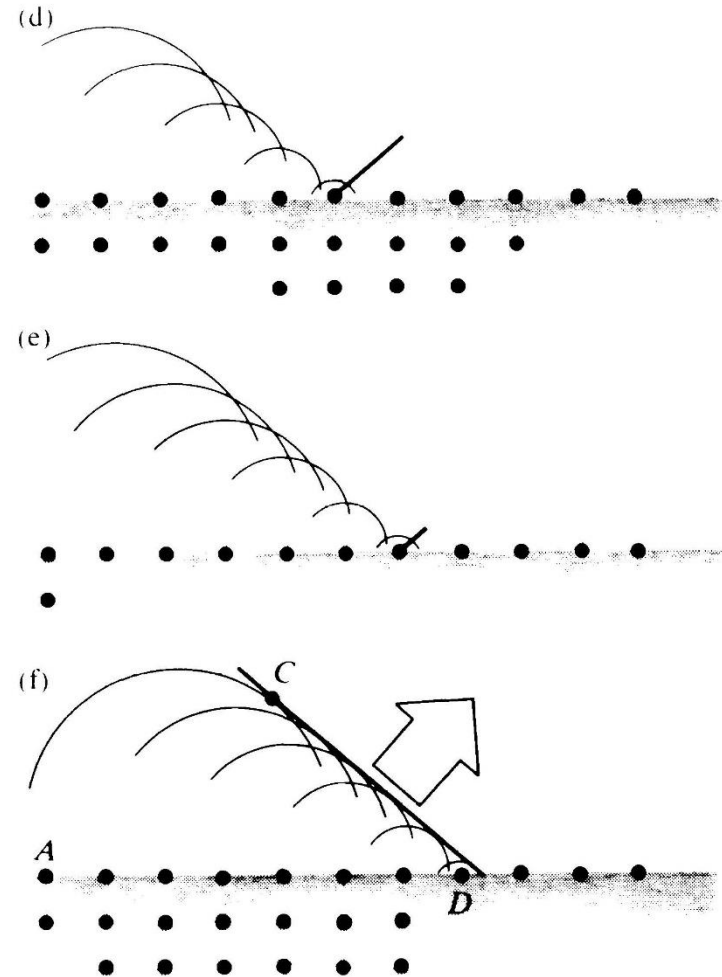
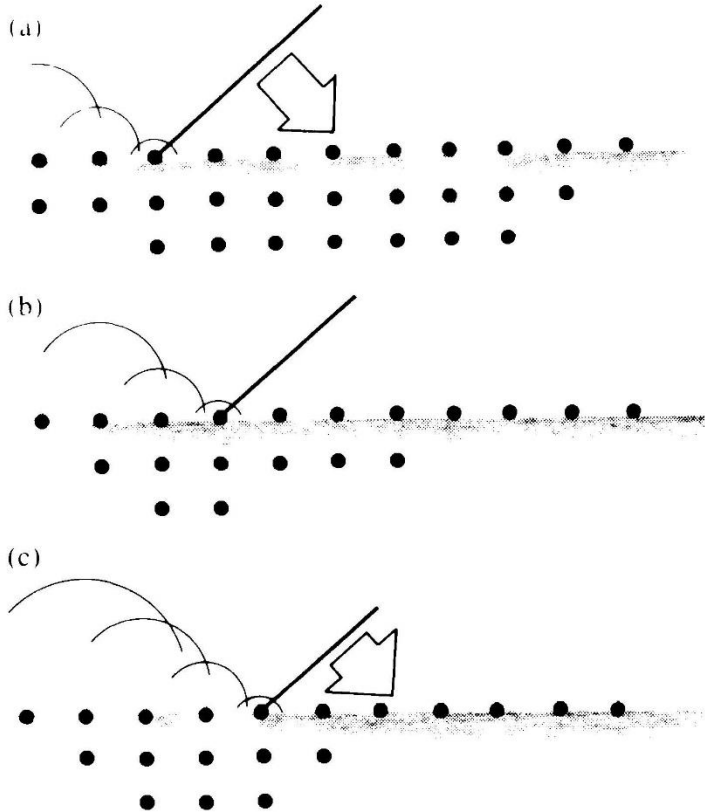


The law of reflection



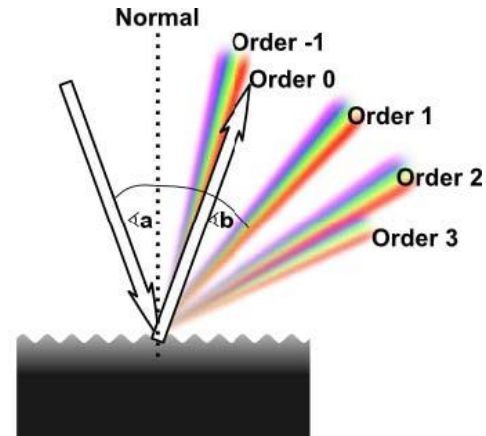
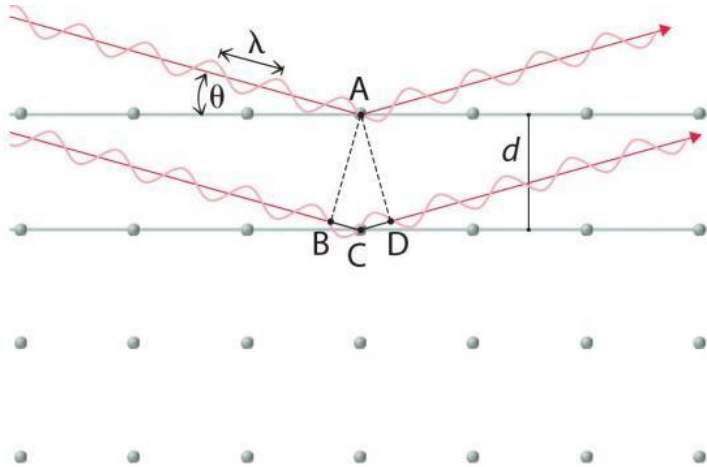
Wavelength $\gg$ separation (molecules or atoms):
A plane wave sweeps in stimulating atoms across the interface.
These radiate and reradiate, thereby giving rise to both the reflected and transmitted waves

How it happens

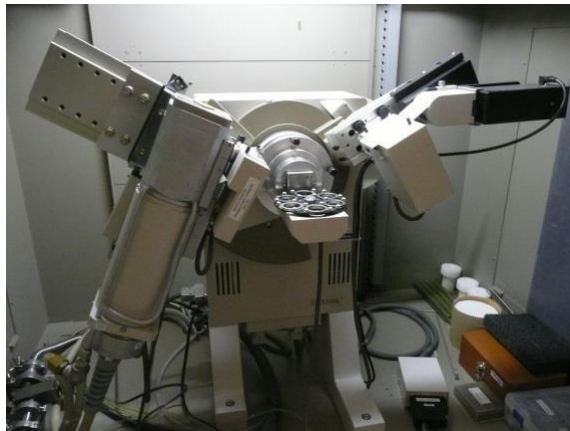


The reflection of a wave as the result of scattering

□ Wavelength < separation: **diffraction**



The X-ray diffractometer

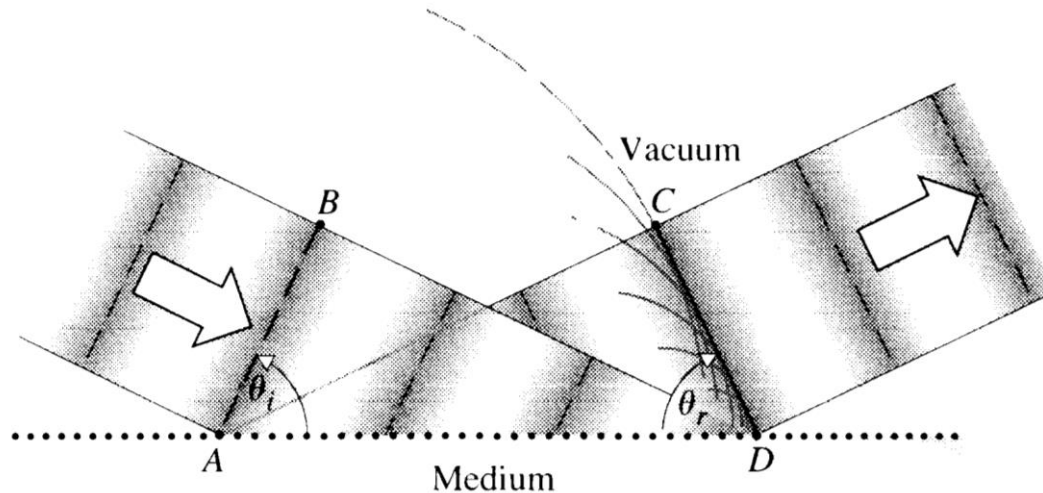


Spectrometer



Law of reflection

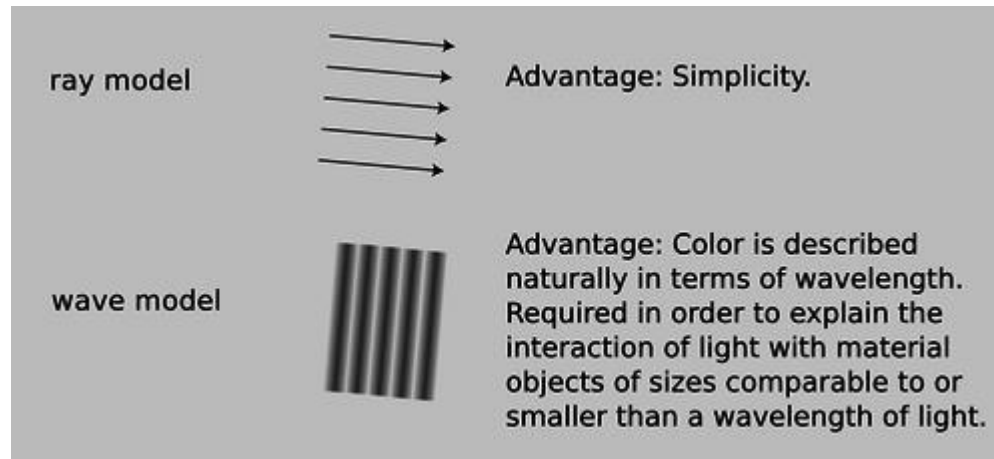
The wavelet emitted from A will arrive at C in-phase with the wavelet just being emitted from D, as long as the distances $\overline{AC}$ and $\overline{BD}$ are equal.



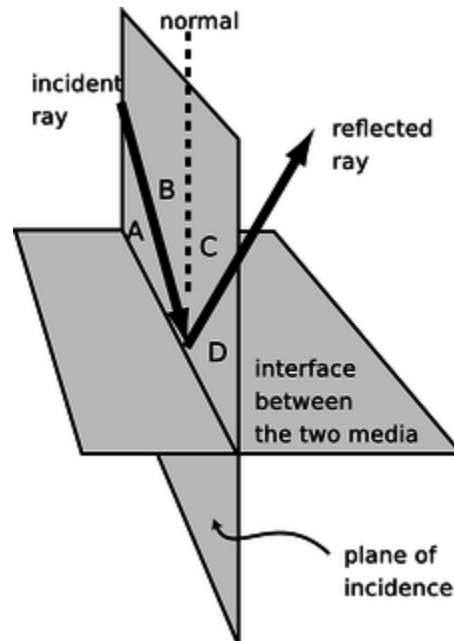
$$\frac{\sin \theta_i}{\overline{BD}} = \frac{\sin \theta_r}{\overline{AC}} \quad \overline{BD} = v_i \Delta t = \overline{AC}$$

$\theta_i = \theta_r$ ❖ The angle-of-incidence equals the angle-of-reflection

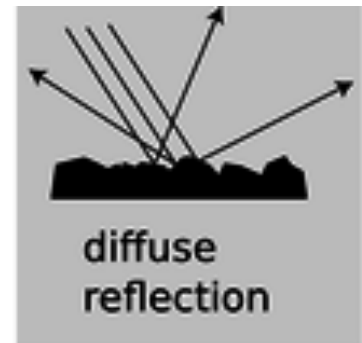
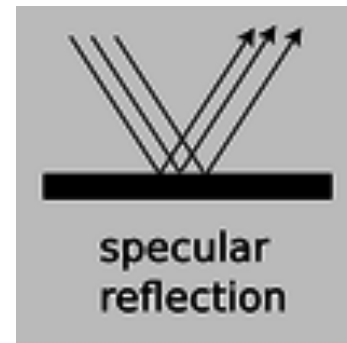
Ray



Plane-of-incidence: the plane where the incident ray, the perpendicular to the surface, and the reflected ray all lie in.



Specular and diffuse reflection



3.4 Refraction

Immediately on entering the transmitting medium, the transmitted wave propagates with an effective speed $v_t < c$ for air-glass interface.

The law of refraction:

In the time Δt , which it takes for point B on a wavefront (traveling at speed v_i) to reach point D, the transmitted portion of that same wavefront (traveling at speed v_t) has reached point E.

$$\frac{\sin \theta_i}{\overline{BD}} = \frac{\sin \theta_t}{\overline{AE}}$$

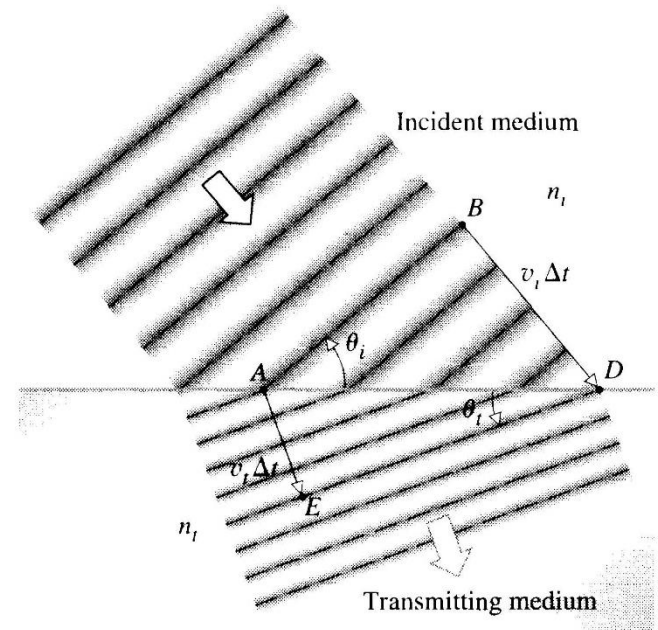
$$\overline{BD} = v_i \Delta t$$

$$\overline{AE} = v_t \Delta t$$

$$\frac{\sin \theta_i}{v_i} = \frac{\sin \theta_t}{v_t}$$

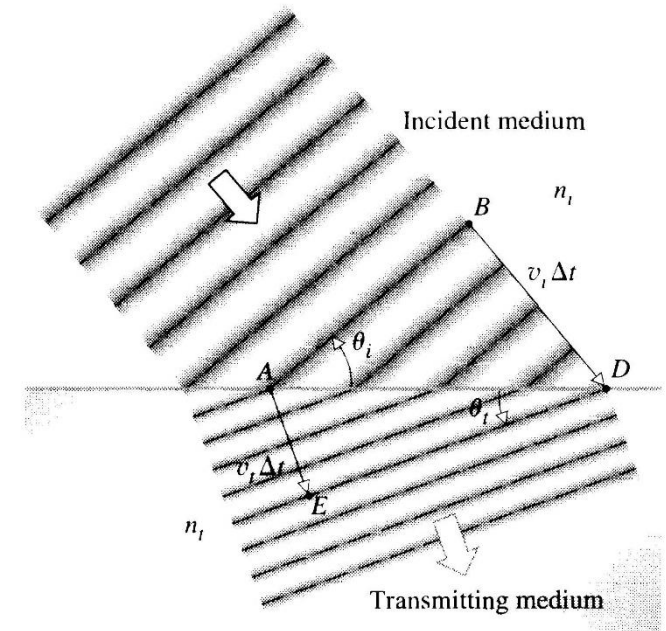
$$n_i \sin \theta_i = n_t \sin \theta_t$$

The law of refraction or Snell's law



Three important changes that occur in the beam traversing the interface:

- ❑ 1) it changes direction;
- ❑ 2) the beam in the glass has a broader cross section than the beam in the air;
- ❑ 3) the wavelength decrease because the frequency is unchanged with the speed decreases.

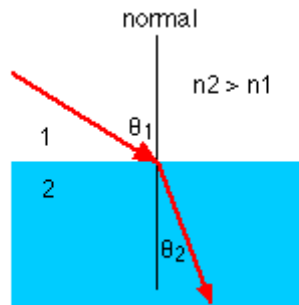
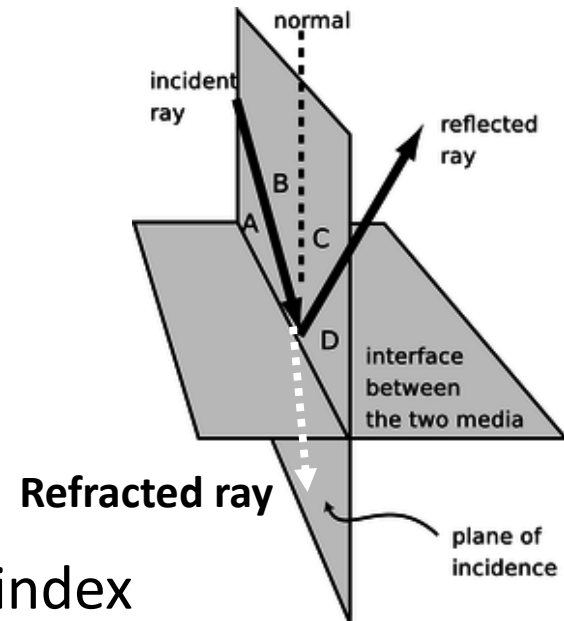


Note: the color aspect of light is better thought of as associated with its frequency (or energy, $E=h\nu$)

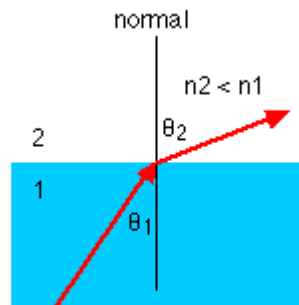
When we talk about wavelength and colors, we should always be referring to **vacuum wavelengths**.

$$\lambda = \frac{v}{\nu} = \frac{c}{n\nu} = \frac{\lambda_0}{n}$$

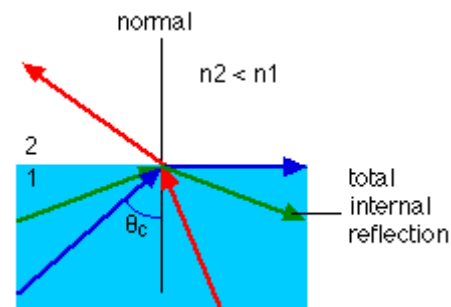
The incident, reflected, and refracted rays all lie in the plane-of-incidence



The ray entering a higher-index medium bends toward the normal.



On entering a medium having a lower index, the ray will bend away from the normal.

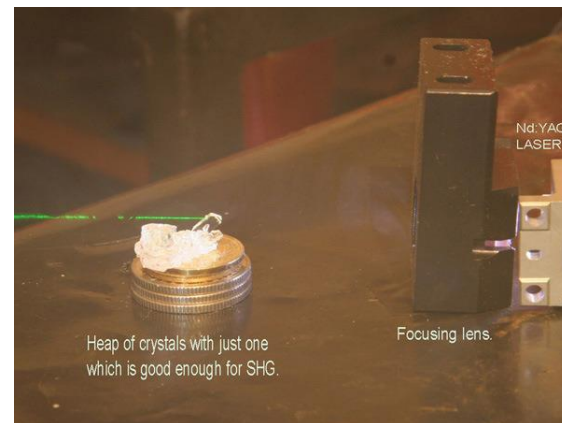


Total internal reflection

More

- ❑ In all the situations treated thus far, it was assumed that the reflected and refracted beams always had **the same frequency** as the incident beam.
- ❑ It is the truth when the amplitude of the vibration is fairly small.
- ✓ E-field for bright sunlight: 1000V/m
- ✓ Cohesive field holding the electron in an atom: 10^{11}V/m .
- ❑ It is not true for exceedingly large-amplitude E-field, for example high power laser (pulse laser)

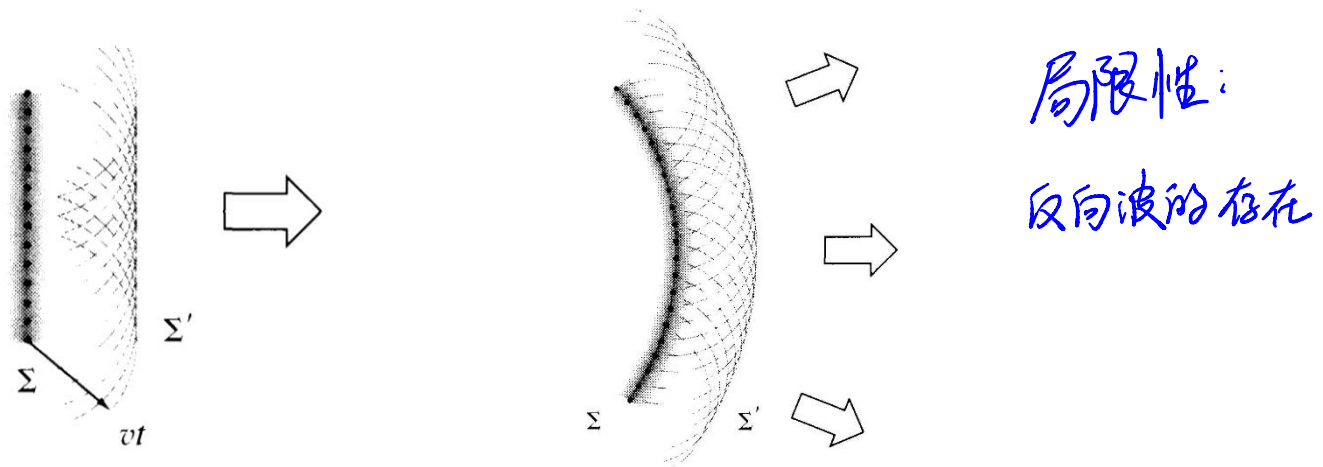
Nonlinear effect: for example
second harmonic generation (SHG)



3.4.2 Huygens' Principle

Every point on a propagating wavefront serves as the source of spherical secondary wavelets, such that the wavefront at some later time is the envelope of these wavelets.

The wavelets themselves are a theoretical construct.

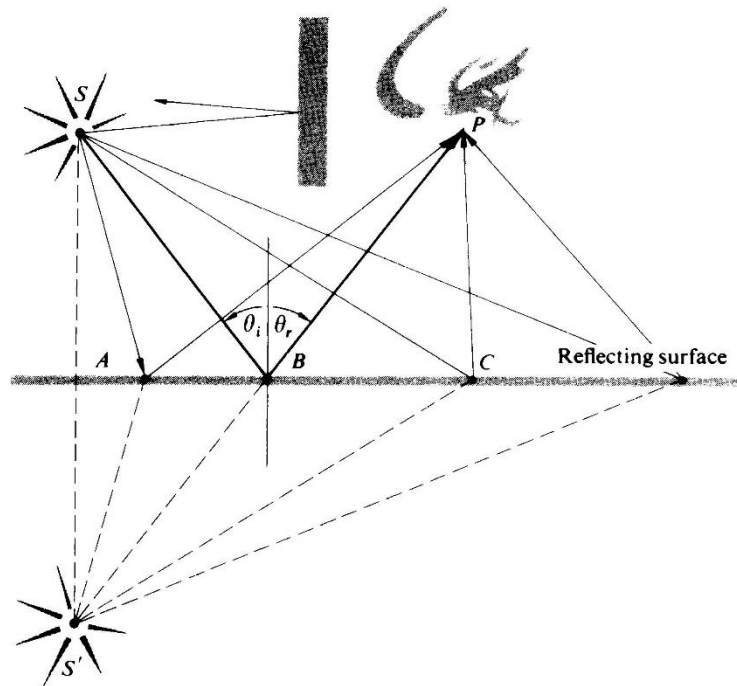


Huygens-Fresnel Principle: Fresnel successfully modified Huygens' Principle mathematically adding in the concept of interference.

费马原理.

3.5 Fermat's principle

Hero's statement: "the path taken by light in going from some point S to a point P via a reflecting surface was the shortest possible one"



Fermat consequently reformulated Hero's statement to read: The actual path between two points taken by a beam of light is the one that is traversed in the least time.

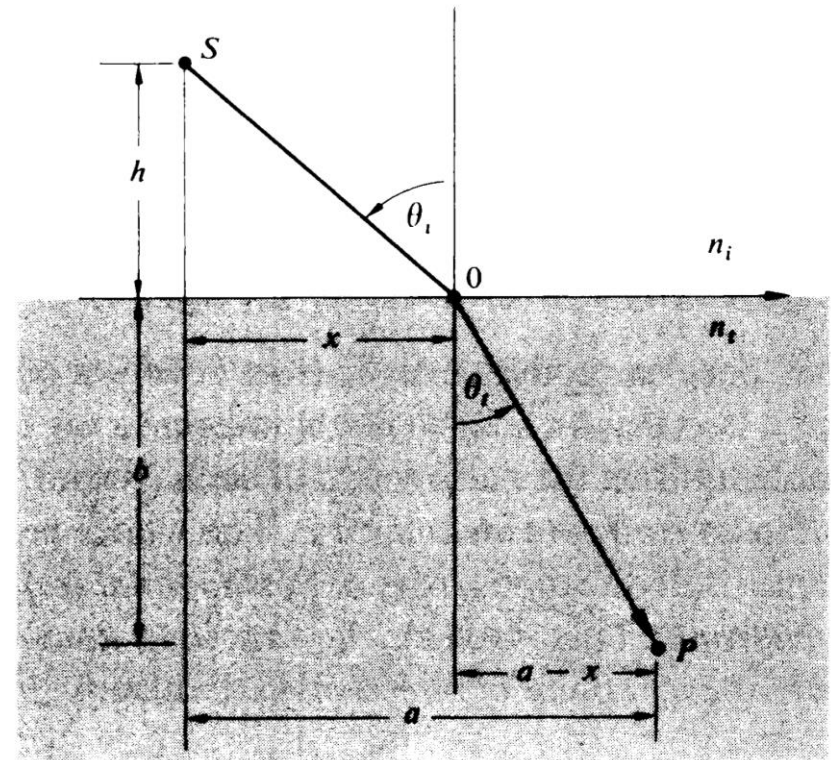
According to Fermat's principle:

$$t = \frac{\overline{SO}}{v_i} + \frac{\overline{OP}}{v_t}$$

The solution:

$$\frac{\sin \theta_i}{v_i} = \frac{\sin \theta_t}{v_t}$$

The law of refraction



Optical path length

The transit time from S to P:

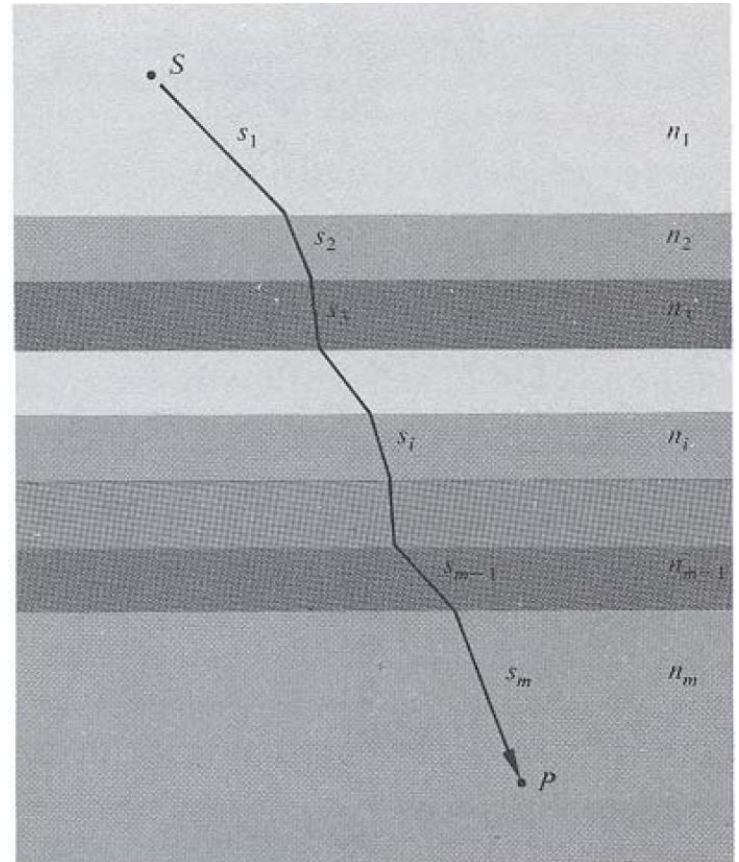
$$t = \frac{s_1}{v_1} + \frac{s_2}{v_2} + \dots + \frac{s_m}{v_m}$$

or

$$t = \sum_{i=1}^m \frac{s_i}{v_i} = \frac{1}{c} \sum_{i=1}^m n_i s_i$$

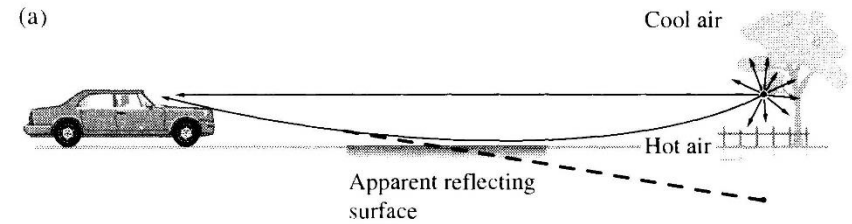
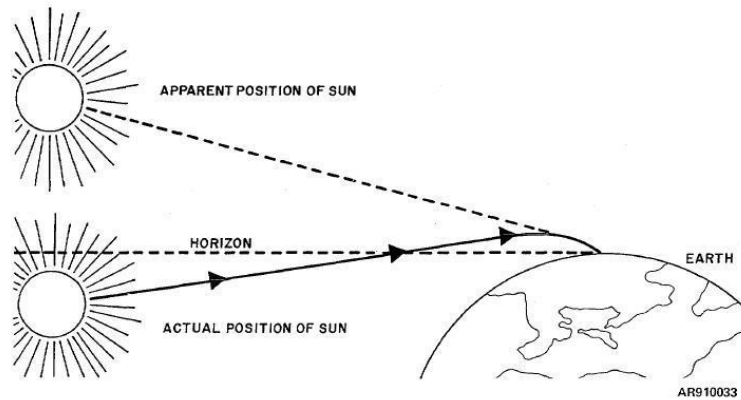
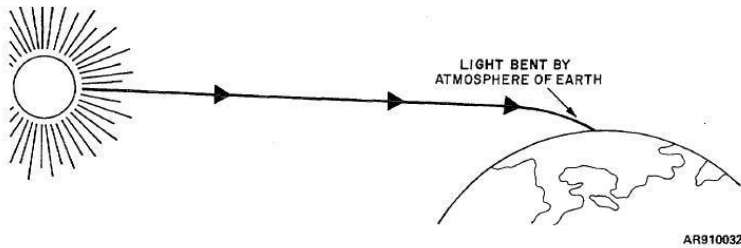
For an inhomogeneous medium,
the **optical path length**:

$$OPL = \int_S^P n(s) ds$$



- ❑ It corresponds to the distance in vacuum equivalent to the distance traversed (s) in the medium of index n .
- ❑ Fermat's Principle: light, in going from point S to P, traverses the route having the smallest optical length.

Fermat and mirages



气温↑, 折射率↑

3.6 The electromagnetic approach

- ❑ As to reflection and refraction phenomena:
 - ✓ Scattering theory
 - ✓ Fermat's principle
- ❑ Another powerful approach:
 - ✓ **Electromagnetic theory**

3.6.1 Waves at an interface

Suppose that the incident monochromatic lightwave is planar:

入射的

单色的

平面的

$$\vec{E}_i = \vec{E}_{0i} \cos(\vec{k}_i \cdot \vec{r} - \omega_i t)$$

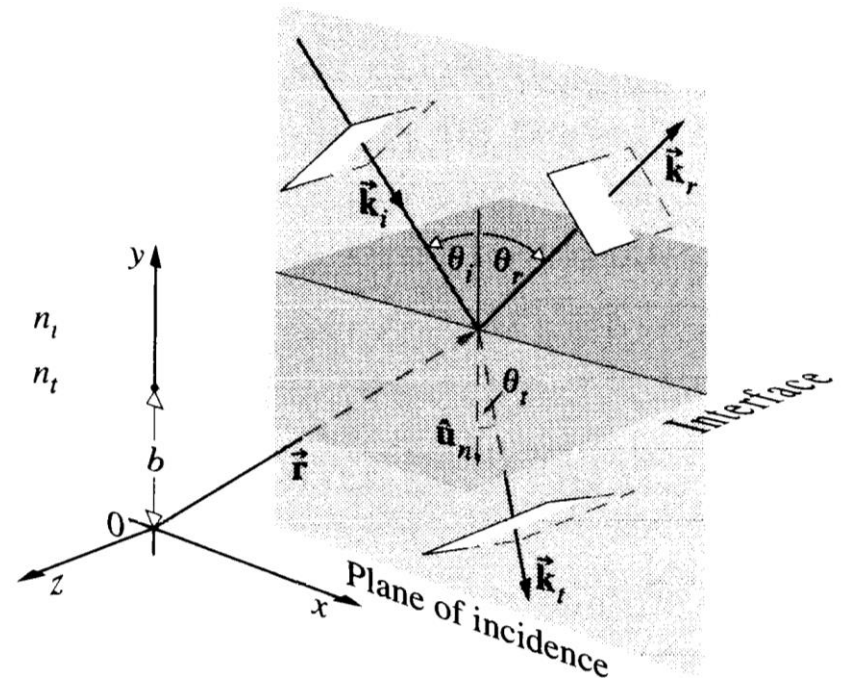
Reflected lightwave:

$$\vec{E}_r = \vec{E}_{0r} \cos(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r)$$

Refracted lightwave:

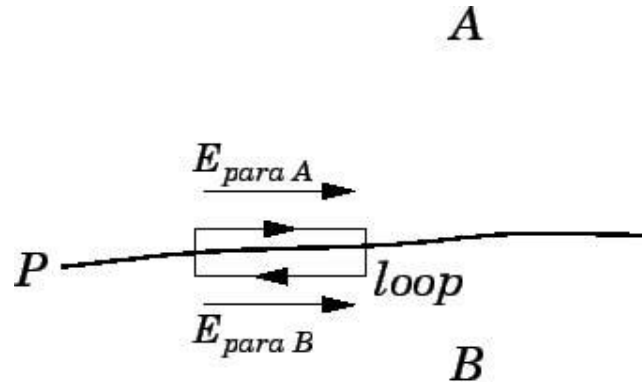
$$\vec{E}_t = \vec{E}_{0t} \cos(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t)$$

ϵ_r and ϵ_t : phase constants relative to E_i



Start from Faraday's induction law:

$$\oint_C \vec{E} \cdot d\vec{\ell} = - \iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

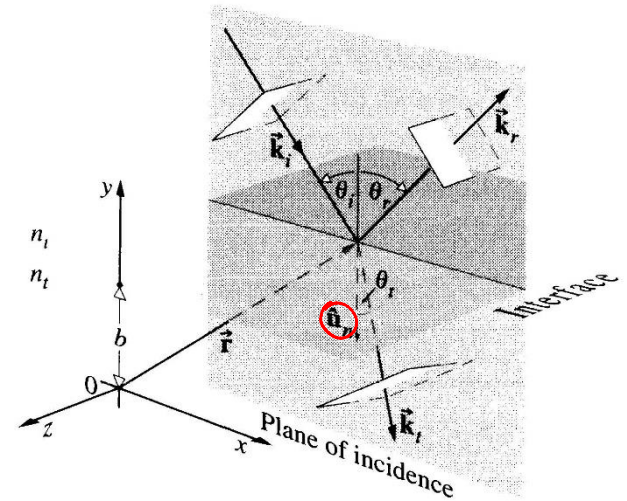


$$E_{\parallel A} = E_{\parallel B}$$

□ Conclusion (Boundary conditions): continuity of tangential E

Boundary condition: continuity of tangential E

$\hat{u}_n$: the unit vector normal to the interface 界面



$$\begin{aligned}\hat{u}_n \times \vec{E}_i + \hat{u}_n \times \vec{E}_r &= \hat{u}_n \times \vec{E}_t \\ \hat{u}_n \times \vec{E}_{0i} \cos(\vec{k}_i \cdot \vec{r} - \omega_i t) \\ + \hat{u}_n \times \vec{E}_{0r} \cos(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r) \\ &= \hat{u}_n \times \vec{E}_{0t} \cos(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t)\end{aligned}$$

This relationship must obtain at any instant in time and at any point on the interface. Three electric fields have the same functional dependence on the t and r .

$$\left(\vec{k}_i \cdot \vec{r} - \omega_i t \right) \Big|_{y=b} = \left(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r \right) \Big|_{y=b} = \left(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t \right) \Big|_{y=b}$$

The frequency is unchanged under linear region

$$\left(\vec{k}_i \cdot \vec{r} \right) \Big|_{y=b} = \left(\vec{k}_r \cdot \vec{r} + \epsilon_r \right) \Big|_{y=b} = \left(\vec{k}_t \cdot \vec{r} + \epsilon_t \right) \Big|_{y=b}$$

$$\left(\vec{k}_i \cdot \vec{r} \right) \Big|_{y=b} = \left(\vec{k}_r \cdot \vec{r} + \epsilon_r \right) \Big|_{y=b}$$

$$\left[\left(\vec{k}_i - \vec{k}_r \right) \cdot \vec{r} \right] \Big|_{y=b} = \epsilon_r$$

The endpoint of r sweeps out a plane perpendicular to the vector $\vec{k}_i - \vec{k}_r$

And the vector $\vec{k}_i - \vec{k}_r$ is parallel to $\hat{u}_n$

$$\hat{u}_n \times (\vec{k}_i - \vec{k}_r) = 0$$

The law of reflection

$$k_i \sin \theta_i = k_i \sin \theta_r$$

$$\theta_i = \theta_r$$

$$\left(\vec{k}_i \cdot \vec{r} \right) \Big|_{y=b} = \left(\vec{k}_t \cdot \vec{r} + \epsilon_t \right) \Big|_{y=b}$$

Snell's law

斯涅尔定律

similarly $k_i \sin \theta_i = k_t \sin \theta_t$

$$\underline{n_i \sin \theta_i = n_t \sin \theta_t}$$

Boundary condition: continuity of tangential H

切向 H

Ampere-Maxwell's Law

安培-麦克斯韦定律

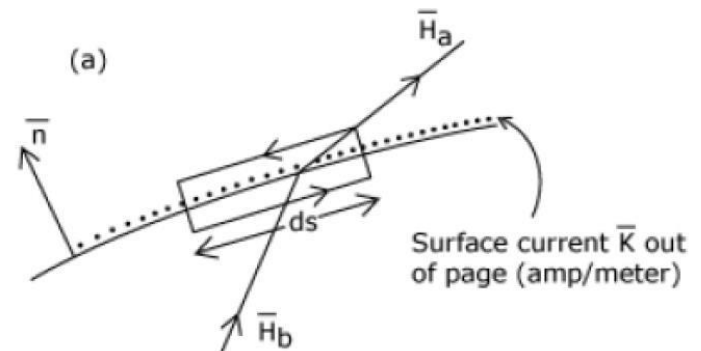
$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

Ampere-Maxwell's Law in matter

$$\oint_C \vec{H} \cdot d\vec{\ell} = \iint_A \left(\vec{J}_f + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$$

$$\vec{H} = \vec{B} / \mu$$

$$\vec{D} = \epsilon \vec{E}$$



无电介质

Dielectric-free space boundary condition

$$\underline{H_{\parallel A} = H_{\parallel B}}$$

3.6.2 The Fresnel Equations

菲涅尔

Case 1: $\vec{E}$ perpendicular to the plane-of-incidence

$$\vec{k} \times \vec{E} = v\vec{B}$$

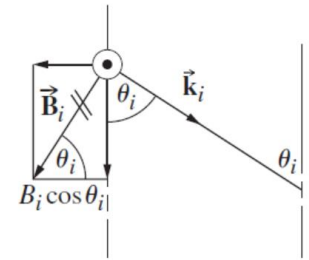
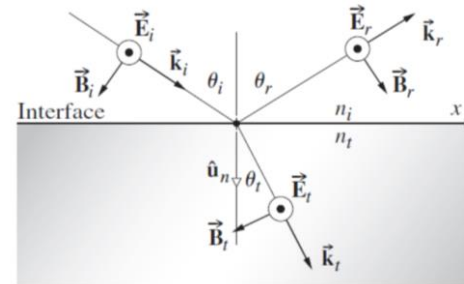
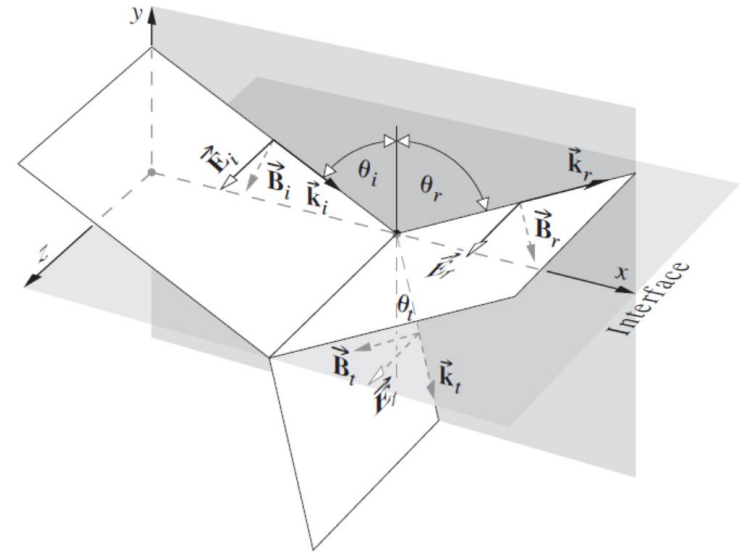
$$\vec{k} \cdot \vec{E} = 0$$

Continuity of the tangential components of the $\vec{E}$ -field at the boundary

$$E_{0i} + E_{0r} = E_{0t}$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$-\frac{B_i}{\mu_i} \cos \theta_i + \frac{B_r}{\mu_i} \cos \theta_r = -\frac{B_t}{\mu_t} \cos \theta_t$$



Reflection and refraction of an incoming wave whose $\vec{E}$ -field is normal to the plane of incidence.

Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Amplitude transmission coefficient:

$$t_{\perp} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\perp} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t}$$

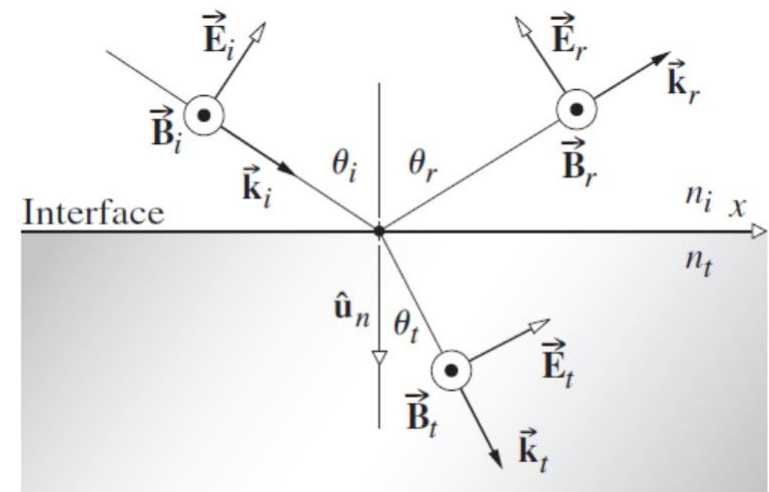
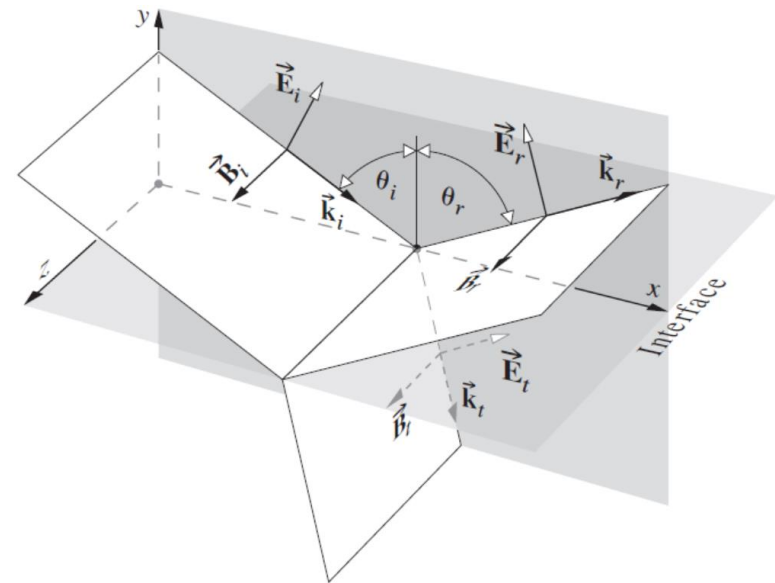
Case 2: $\vec{E}$ parallel to the plane-of-incidence

Continuity of the tangential components of the $\vec{E}$ -field at the boundary

$$E_{0i} \cos \theta_i - E_{0r} \cos \theta_r = E_{0t} \cos \theta_t$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$\frac{B_i}{\mu_i} + \frac{B_r}{\mu_i} = \frac{B_t}{\mu_t}$$



Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_{\parallel} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i}$$

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Amplitude transmission coefficient:

$$t_{\parallel} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\parallel} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_t + n_t \cos \theta_i}$$

$$t_{\perp} \equiv \left(\frac{E_{0t}}{E_{0i}} \right)_{\perp} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t}$$

Further simplification using Snell's law

(考得不会太多)

$$r_{\parallel} = + \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

$$r_{\perp} = - \frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

$$t_{\parallel} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)}$$

$$t_{\perp} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)}$$

Homework

Derive **Fresnel equations**

3.6.3 Interpretation of the Fresnel equations

Amplitude coefficients under nearly normal incidence

$$[r_{\parallel}]_{\theta_i=0} = -[r_{\perp}]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t} \right]_{\theta_i=0}$$

布儒斯特角

$$[t_{\parallel}]_{\theta_i=0} = [t_{\perp}]_{\theta_i=0} = \left[\frac{2n_i}{n_i + n_t} \right]_{\theta_i=0}$$

Example: at an air ($n_i = 1$)–glass ($n_t = 1.5$) interface

$$[r_{\parallel}]_{\theta_i=0} = -[r_{\perp}]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t} \right]_{\theta_i=0} = 0.2$$

External Reflection ($n_t > n_i$) 光从折射率低向折射率高介质传播

$$r_{\parallel} = + \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

When $\theta_i + \theta_t = 90^\circ$, $r_{\parallel} = 0$

This particular value of the incident angle is referred to as **Polarization angle** (θ_p).

The phase shift 180° happens at θ_p . 布儒斯特偏振角

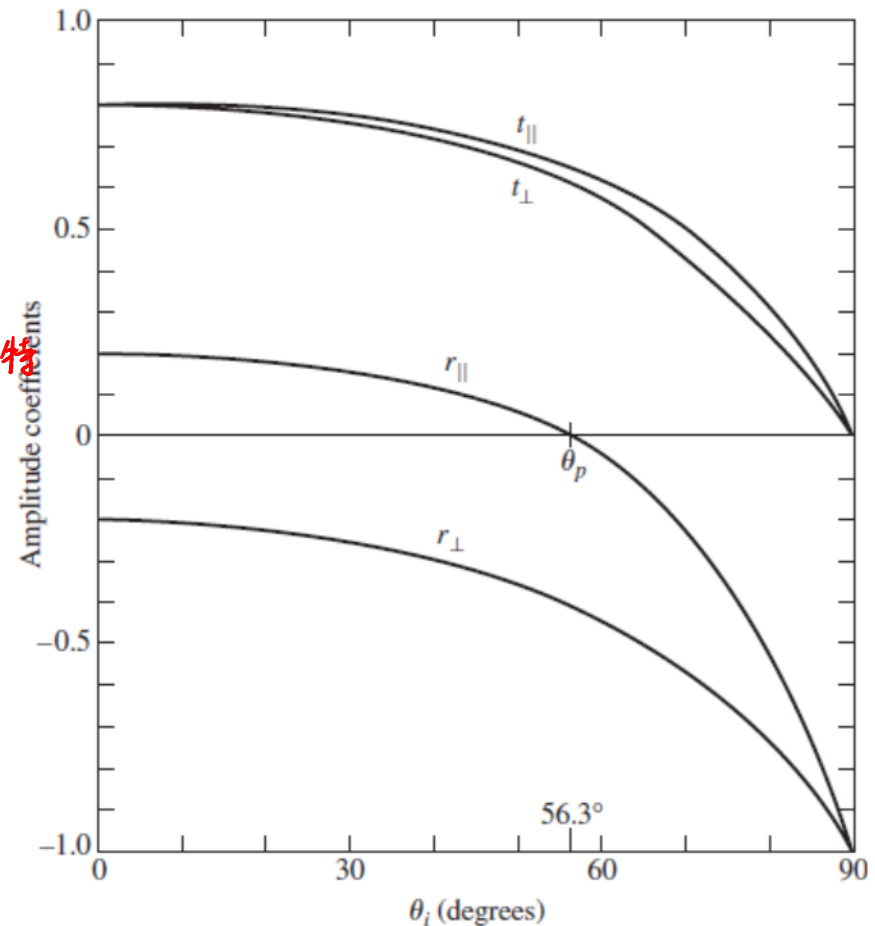
$$r_{\perp} = - \frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} < 0$$

for all incident angle

When $\theta_i = 90^\circ$ (glancing incidence), 掠入射.

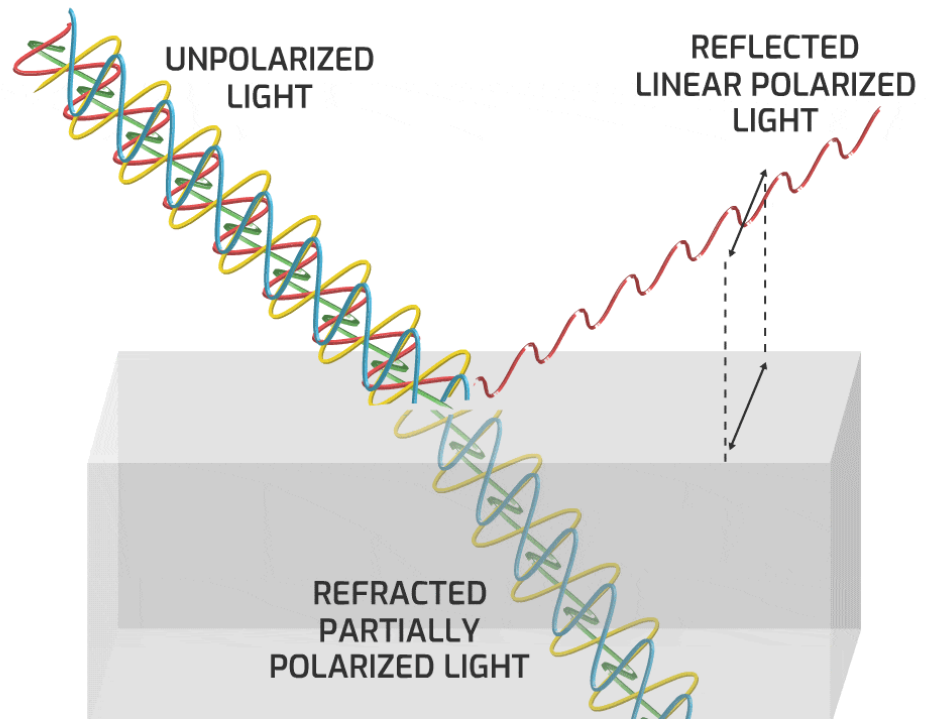
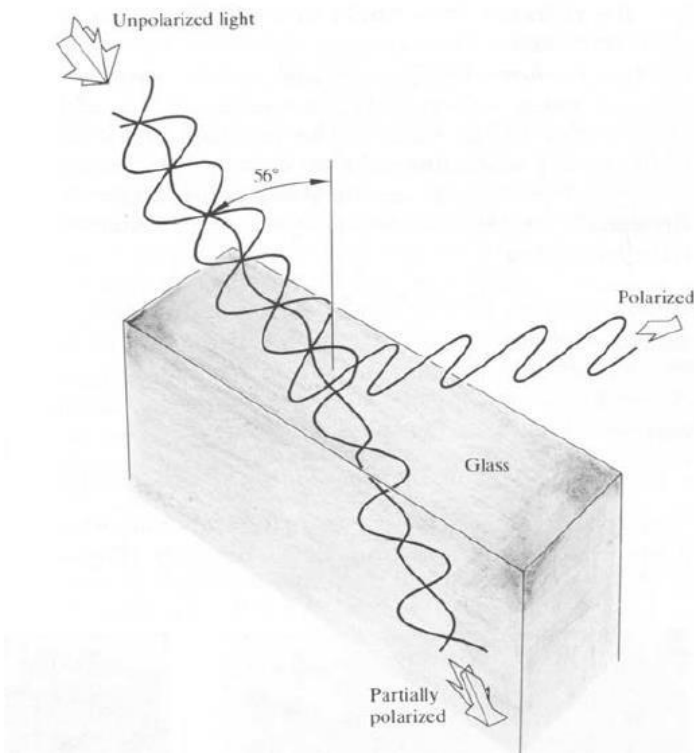
$$r_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i} = -1$$

$$r_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t} = -1$$



The amplitude coefficients of the reflection and transmission as a function of incident angle. These correspond to external reflection $n_t > n_i$ at an air-glass interface.

Polarization angle



Internal Reflection ($n_i > n_t$)

Nearly normal incidence

$$[r_{\parallel}]_{\theta_i=0} = -[r_{\perp}]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t} \right]_{\theta_i=0}$$

With increasing θ_i

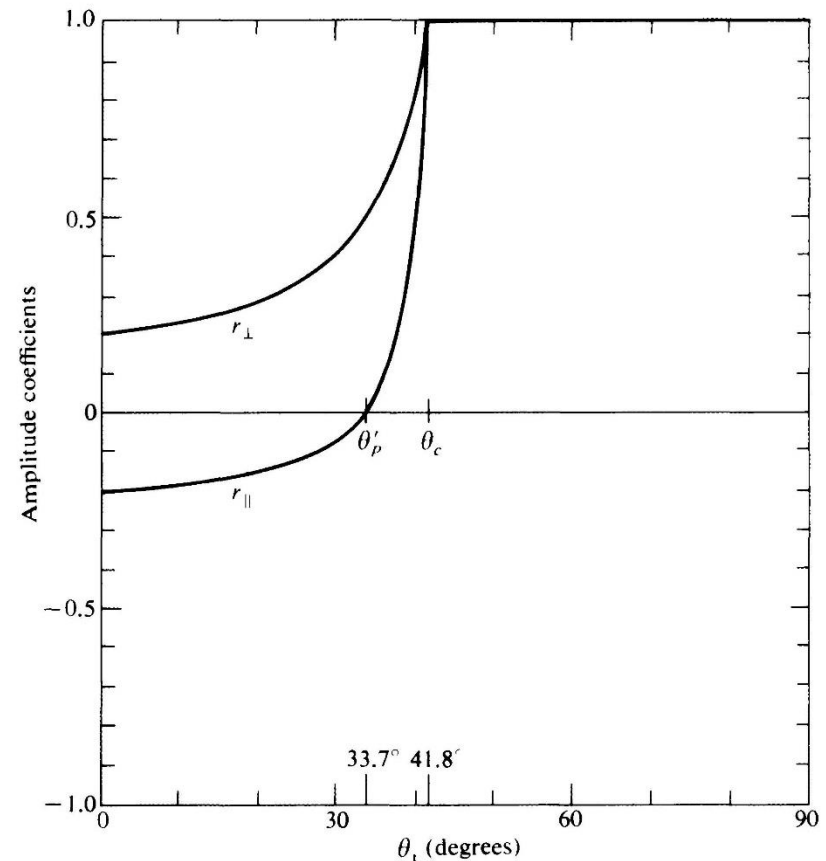
$$r_{\perp} = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} \quad \text{Positive}$$

$$r_{\parallel} = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \quad \begin{array}{l} \text{From negative} \\ \text{to positive at} \\ \theta_i + \theta_t = 90^\circ \end{array}$$

Passing through zero at the polarization angle θ_p'

$$\text{When } \theta_t = 90^\circ \quad r_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i} = 1$$

$$r_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t} = 1$$



The amplitude coefficients of reflection as a function of incident angle. These correspond to internal reflection at an air-glass interface.

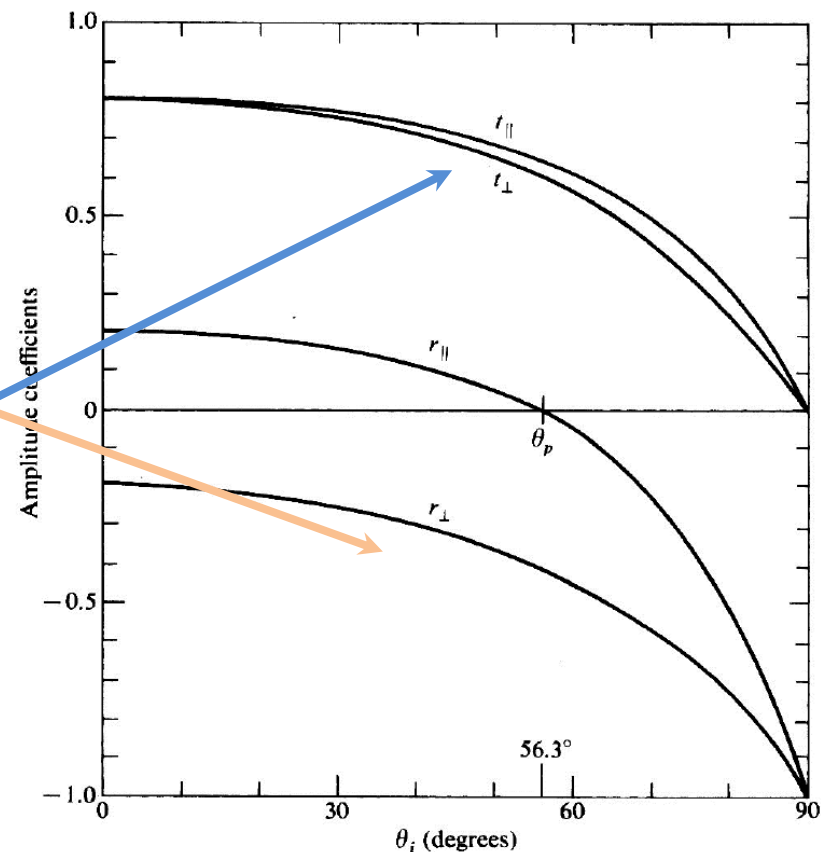
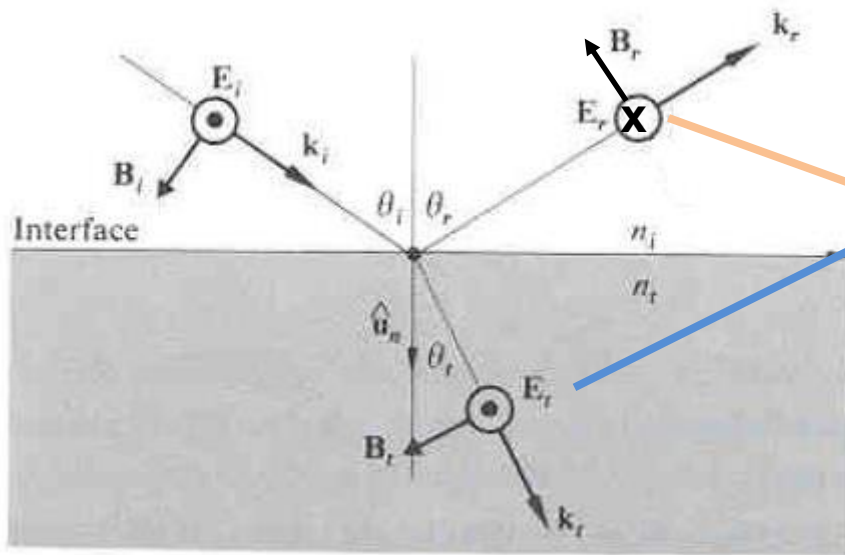
Phase shift for external reflection

$\vec{E}$ perpendicular to the plane-of-incidence

$$r_{\parallel} = + \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \quad r_{\perp} = - \frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

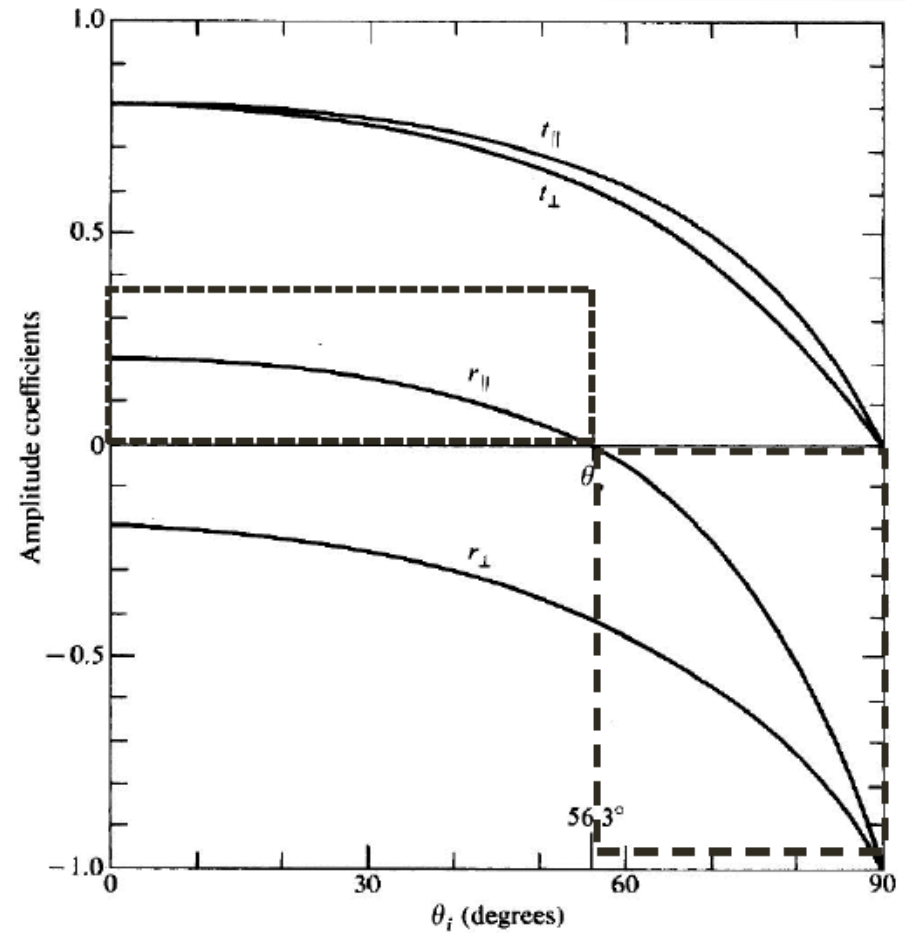
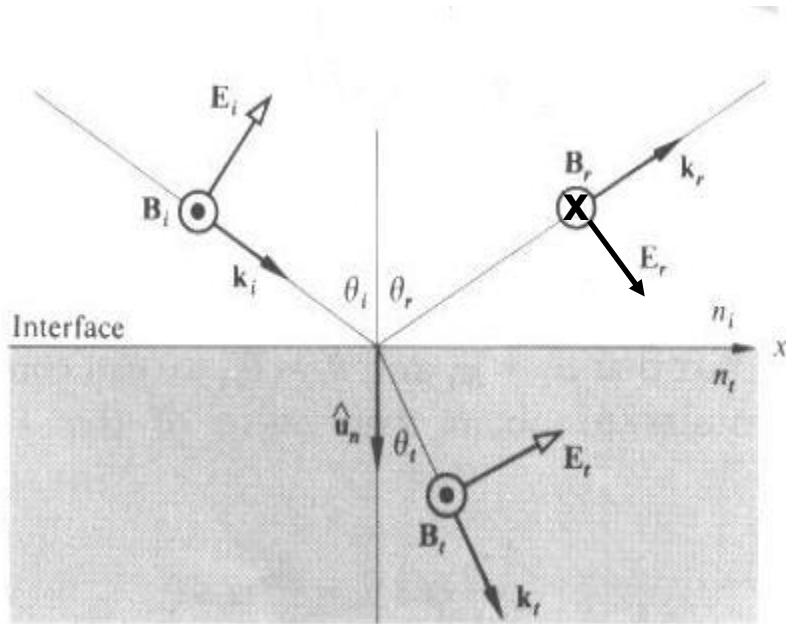
$$t_{\parallel} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t) \cos(\theta_i - \theta_t)}$$

$$t_{\perp} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin(\theta_i + \theta_t)}$$

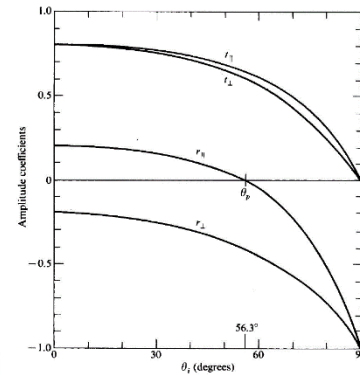
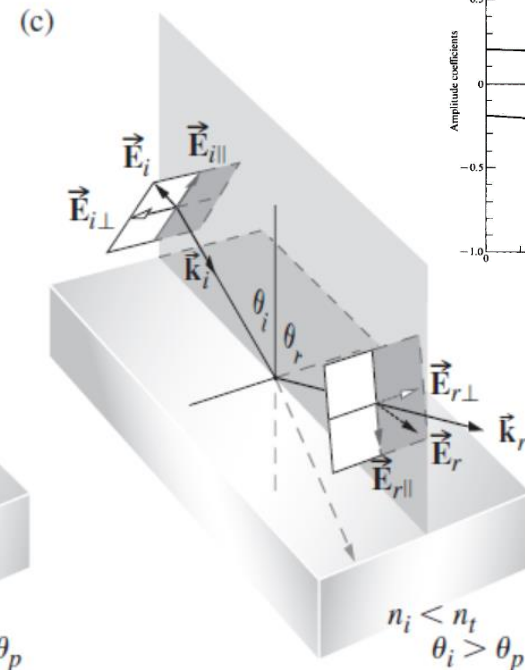
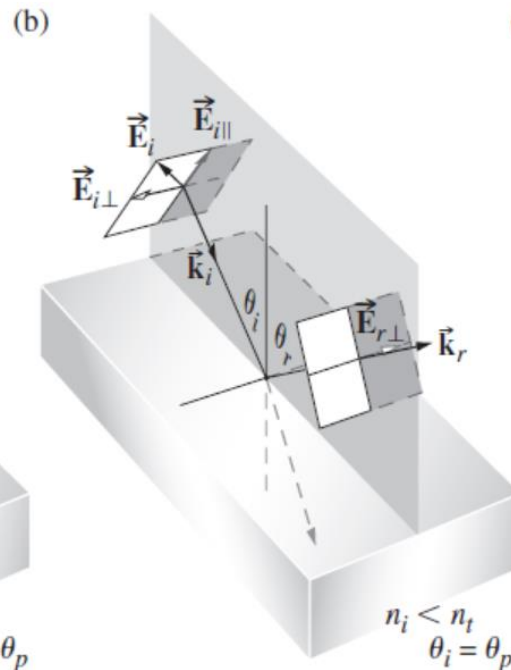
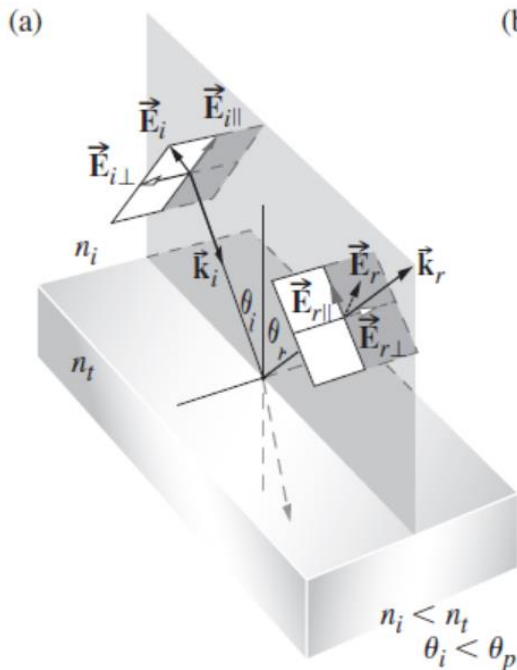
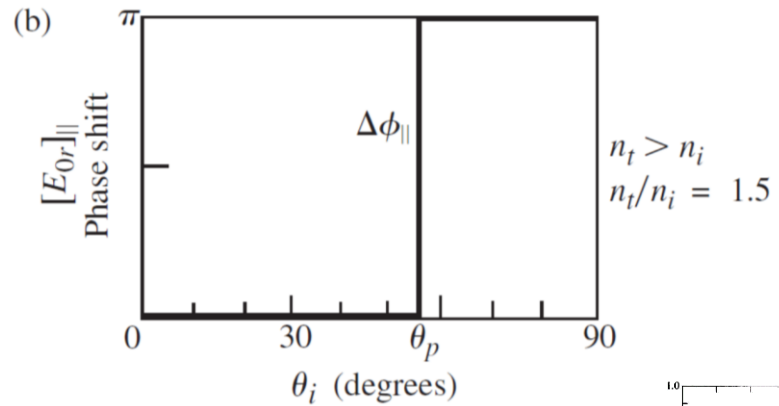
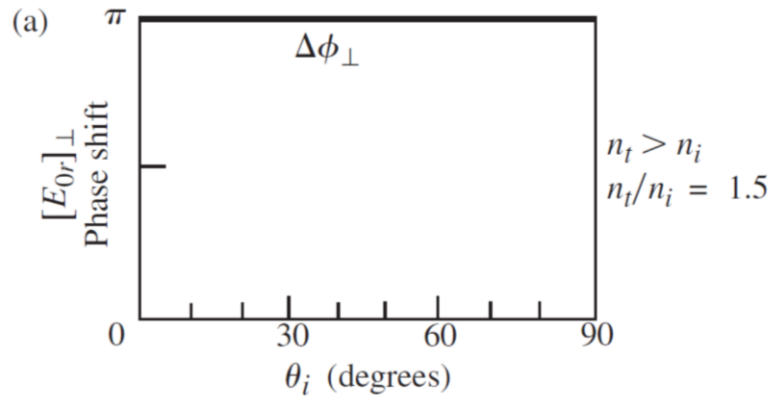


Phase shift for external reflection

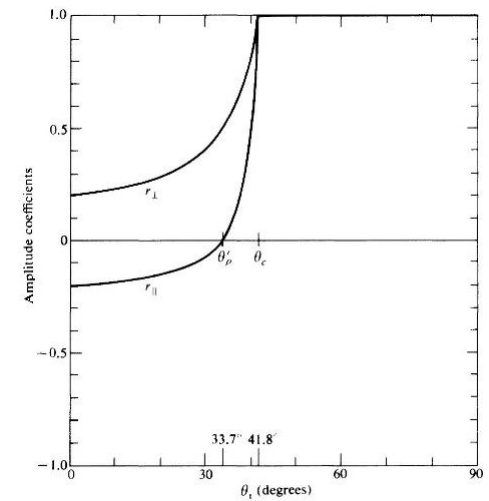
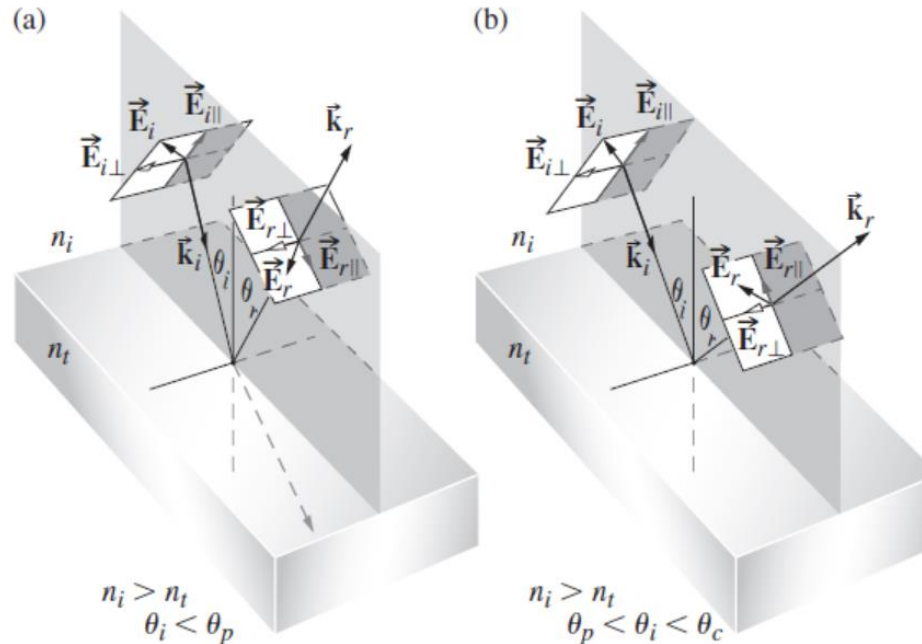
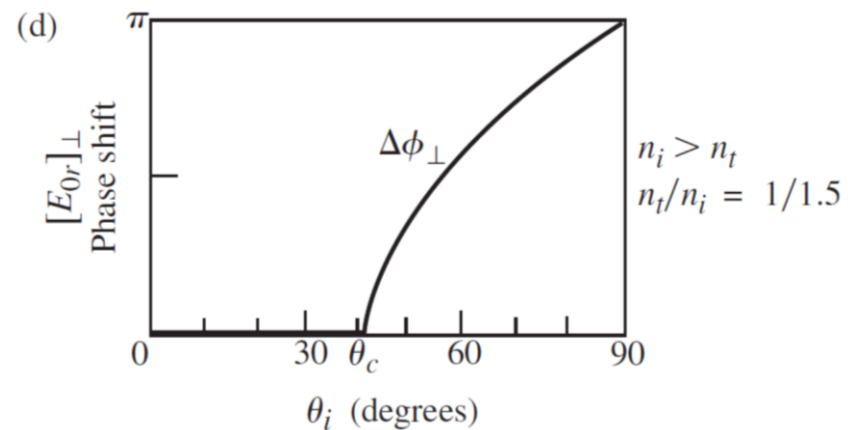
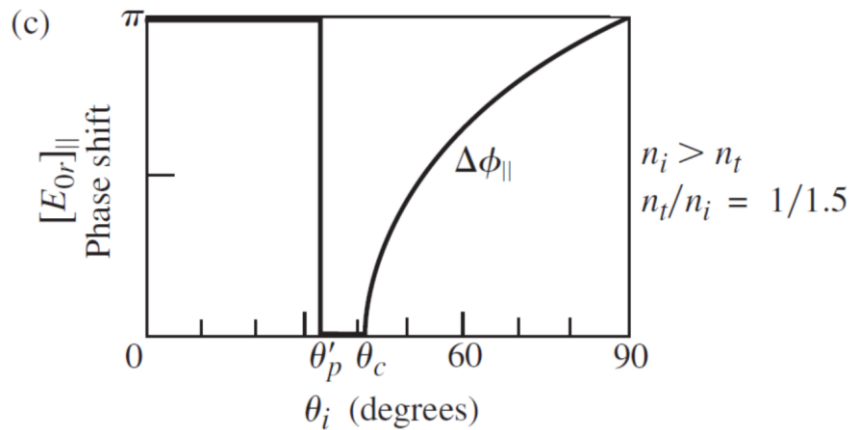
$\vec{E}$ parallel to the plane-of-incidence



Phase shift for external reflection



Phase shift for internal reflection



Reflectance and transmittance

Poynting vector: the power per unit area crossing a surface in vacuum

$$\vec{S} = c^2 \epsilon_0 \vec{E} \times \vec{B}$$

Radiant flux density (W/m²): average energy per unit time crossing a unit area normal to $\vec{S}$.

$$I = \langle S \rangle_T = \frac{c \epsilon_0 E_0^2}{2}$$

Reflectance (R):

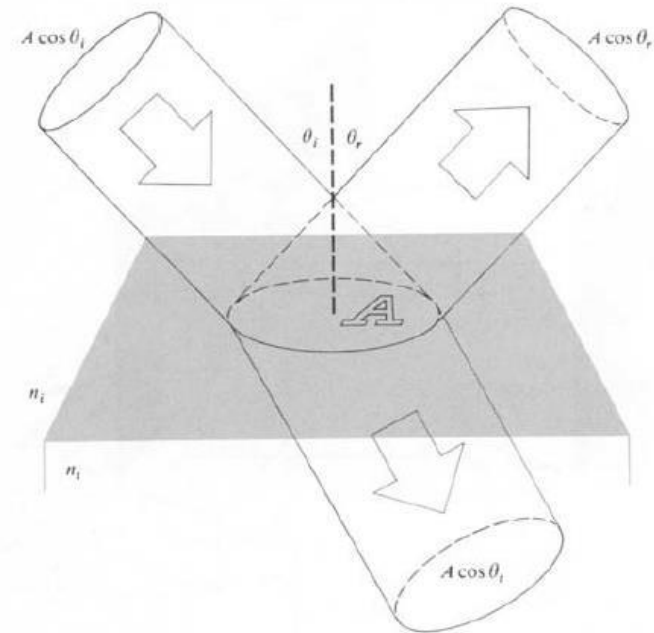
反射率

$$R \equiv \frac{I_r A \cos \theta_r}{I_i A \cos \theta_i} = \frac{I_r}{I_i} = \left(\frac{E_{0r}}{E_{0i}} \right)^2 = r^2$$

Transmittance (T):

透射率

$$T \equiv \frac{I_t A \cos \theta_t}{I_i A \cos \theta_i} = \frac{n_t \cos \theta_t}{n_i \cos \theta_i} \left(\frac{E_{0t}}{E_{0i}} \right)^2 = \left(\frac{n_t \cos \theta_t}{n_i \cos \theta_i} \right) t^2$$



The reason that T is not simply equal to t²: 1) $I \propto \nu$; 2) the cross-sectional areas of the incident and refracted beams are different

Conservation of energy: the total energy flowing into area A per unit time must equal the energy flowing outward from it per unit time

$$I_i A \cos \theta_i = I_r A \cos \theta_r + I_t A \cos \theta_t$$

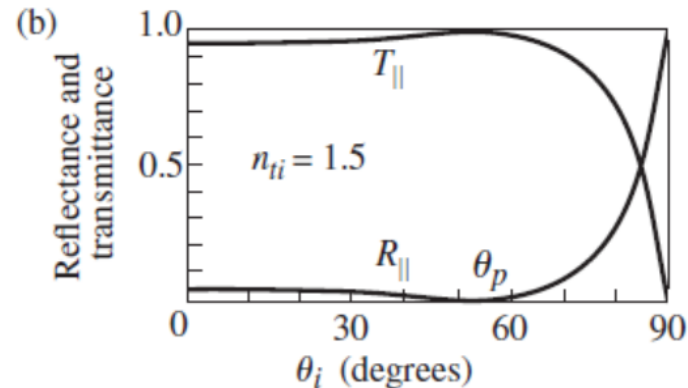
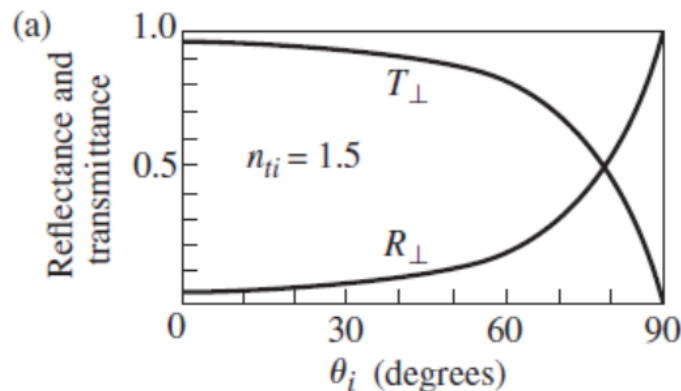


$$R + T = 1$$

The component forms:

$$R_{\perp} = r_{\perp}^2 \quad T_{\perp} = \left(\frac{n_t \cos \theta_t}{n_i \cos \theta_i} \right) t_{\perp}^2$$

$$R_{\parallel} = r_{\parallel}^2 \quad T_{\parallel} = \left(\frac{n_t \cos \theta_t}{n_i \cos \theta_i} \right) t_{\parallel}^2$$



Normal incidence

$$R = R_{\parallel} = R_{\perp} = \left(\frac{n_t - n_i}{n_t + n_i} \right)^2$$

$$T = T_{\parallel} = T_{\perp} = \frac{4n_t n_i}{(n_t + n_i)^2}$$

4% of the light incident normally on an air-glass interface will be reflected back.



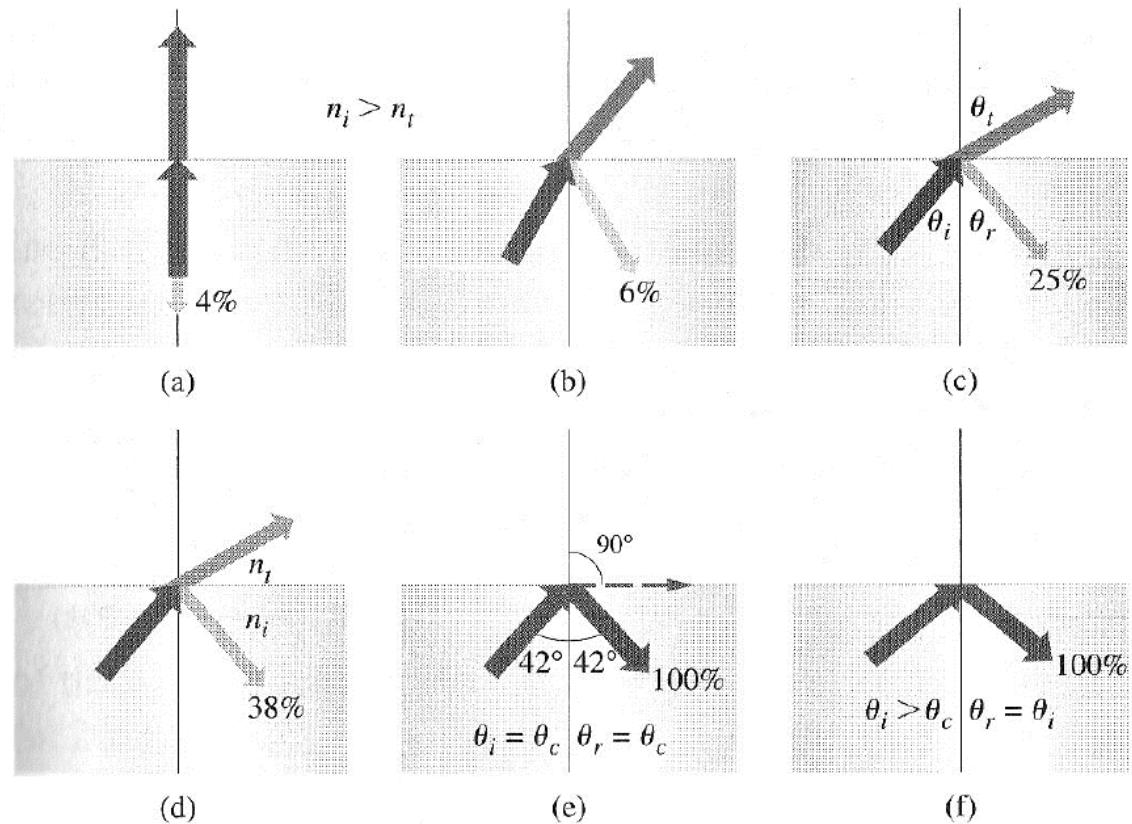
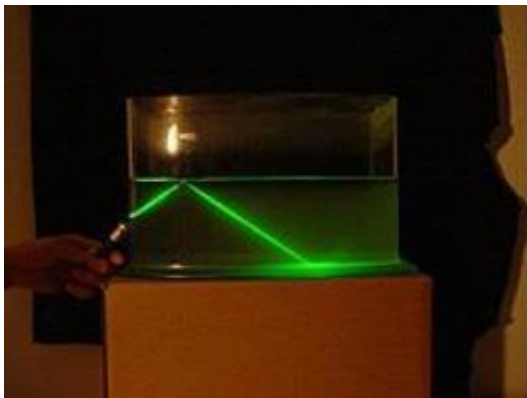
Because it's a lot brighter outside than inside the building,
you have no trouble seeing outside from inside

3.7 Total internal reflection

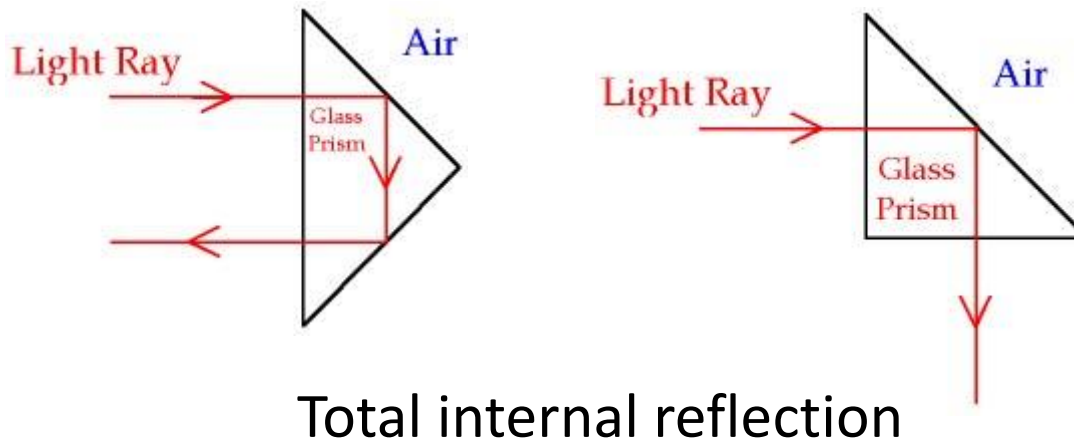
Critical angle: the special value of θ_i for which $\theta_t = 90^\circ$

$$\sin \theta_i = \frac{n_t}{n_i} \sin \theta_t$$

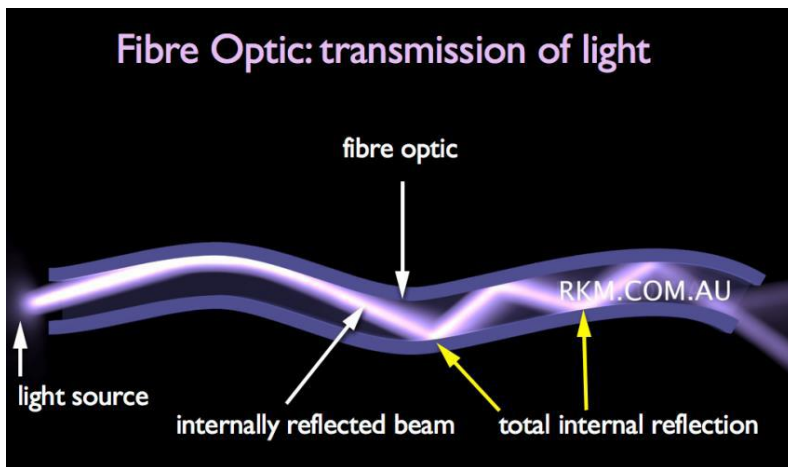
$$\sin \theta_c = \frac{n_t}{n_i} = n_{ti}$$



The critical angle for our air-glass interface is roughly 42° .

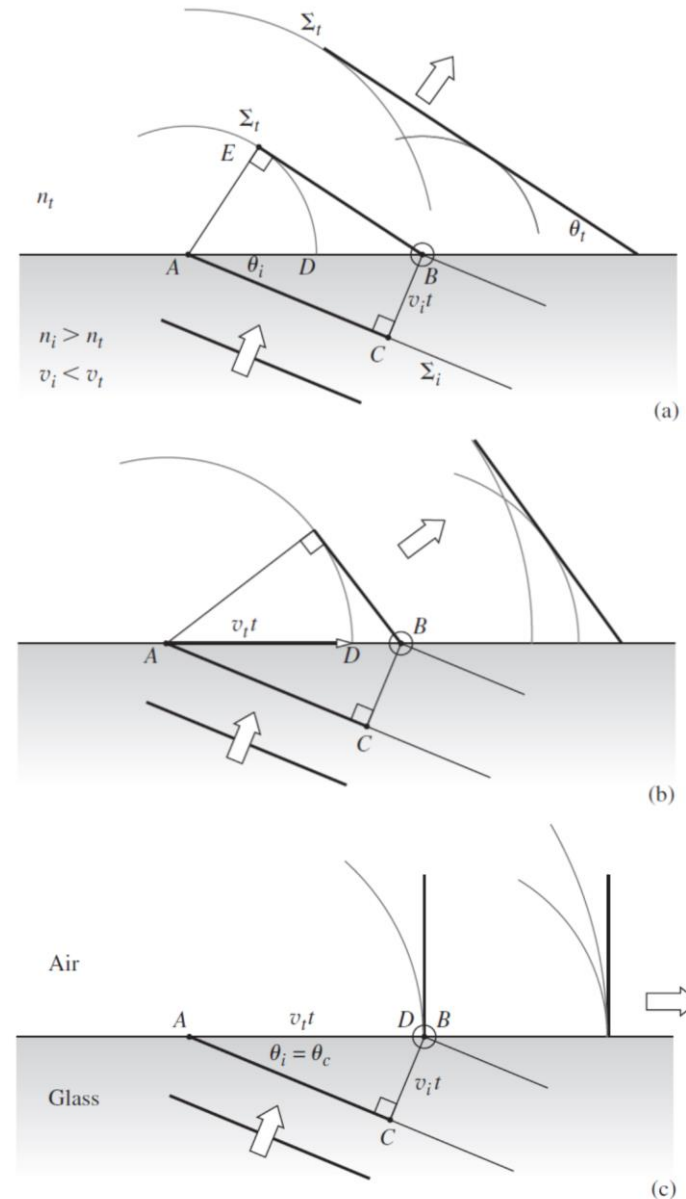


Application



Scattering perspective:

Keep θ_i and n_i constant and in successive parts of the diagram decrease n_t , thereby increasing v_t



Surface wave: for the two given media, the transmitting medium is along the interface. The resulting disturbance is known as a surface wave.

3.7.1 The evanescent wave (倏逝波)

In the case of total internal reflection: no transmitted wave.

But it becomes impossible to satisfy the boundary conditions using only the incident and reflected wave.

$$r_{\parallel} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\parallel} = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_i \cos \theta_t + n_t \cos \theta_i} = \frac{n_{ti}^2 \cos \theta_i - (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}{n_{ti}^2 \cos \theta_i + (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}$$

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}} \right)_{\perp} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t} = \frac{\cos \theta_i - (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}{\cos \theta_i + (n_{ti}^2 - \sin^2 \theta_i)^{1/2}}$$

Since $\sin \theta_c = n_{ti}$, when $\theta_i > \theta_c$, $\sin \theta_i > n_{ti}$, and $r_{\parallel}, r_{\perp}$: complex quantities.

$$r_{\parallel} r_{\parallel}^* = r_{\perp} r_{\perp}^* = 1 \quad R = 1 \quad I_r = I_i \quad \text{and} \quad I_t = 0$$

Although there must be a transmitted wave, it cannot, on average, carry energy across the boundary.

The wavefunction for the transmitted electric field:

$$\vec{E}_t = \vec{E}_{0t} \exp i (\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{k} \cdot \vec{r} = k_{tx}x + k_{ty}y$$

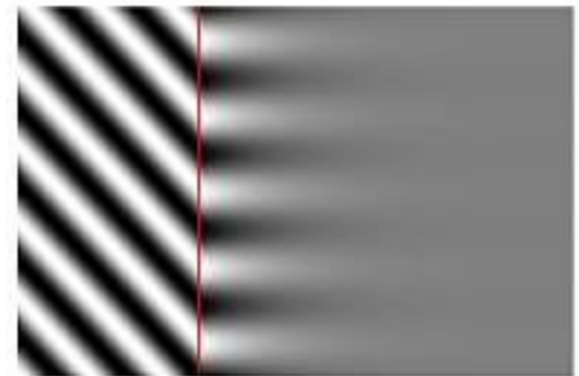
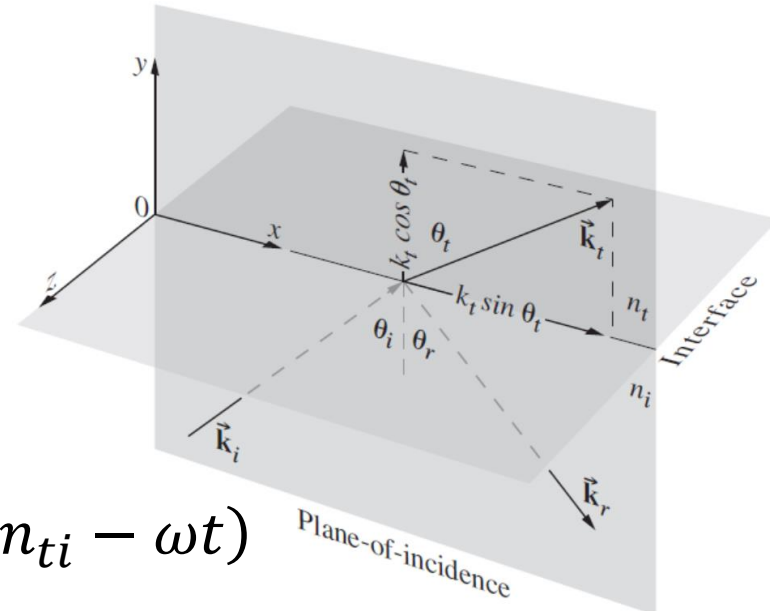
Solution:

$$\vec{E}_t = \vec{E}_{0t} \exp(-\beta y) \exp i (k_t x \sin \theta_i / n_{ti} - \omega t)$$

$$\beta = k_t \left(\frac{\sin^2 \theta_i}{n_{ti}^2} - 1 \right)^{\frac{1}{2}} \quad \delta = \frac{1}{\beta} \quad \text{penetration depth}$$

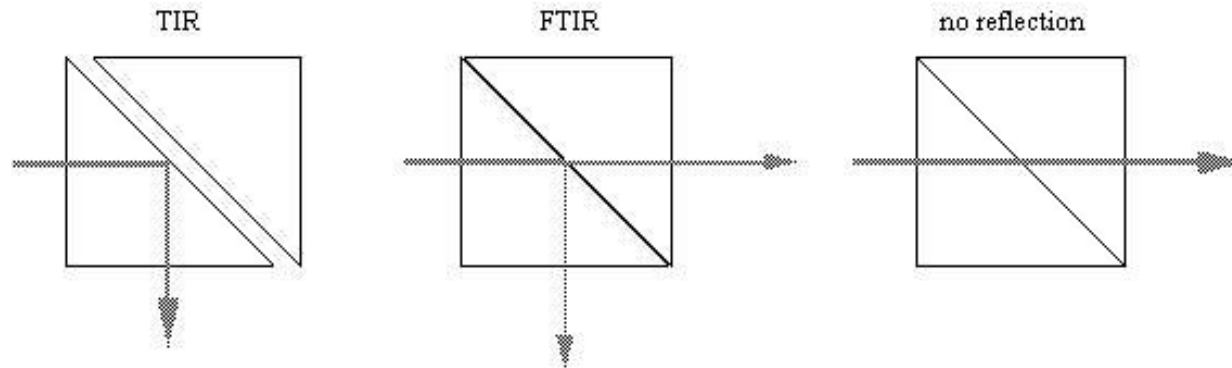
We have a wave whose amplitude drops off exponentially as it penetrates the less dense medium. The disturbance advances in the x-direction as a surface or **evanescent** wave .

Its amplitude decays rapidly in the y-direction, becoming negligible at a distance into the second medium of only a few wavelengths.

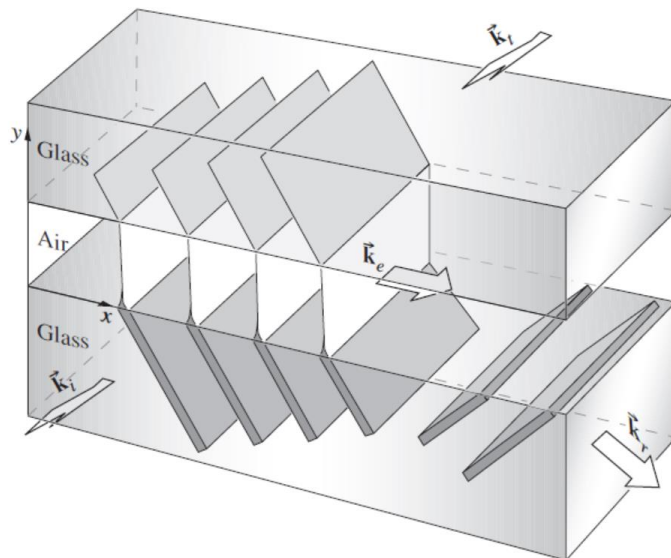


evanescent wave

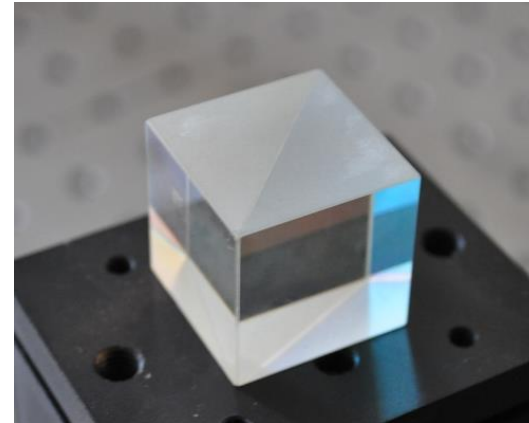
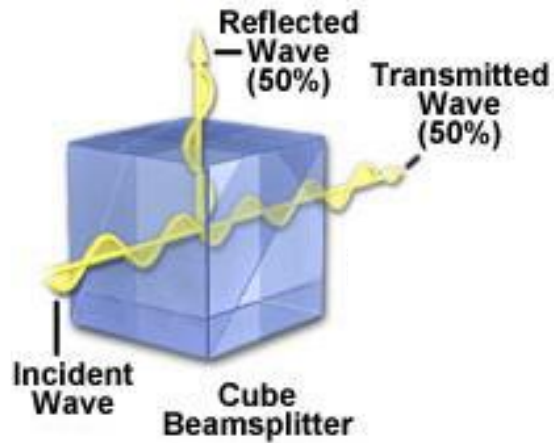
Frustrated total internal reflection (FTIR, 受抑全内反射)



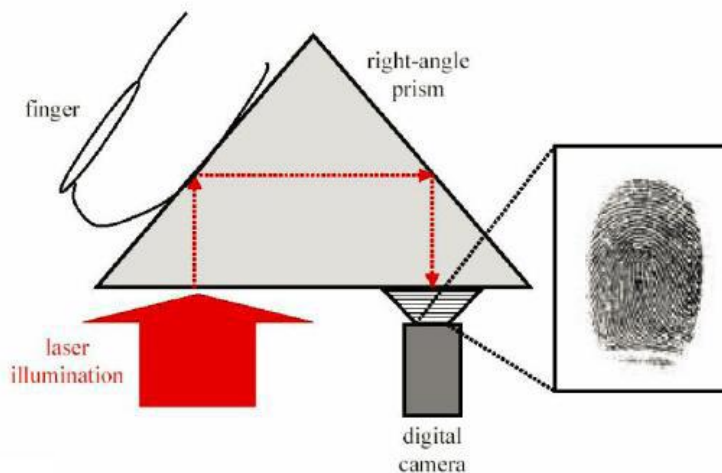
In more general terms, when the evanescent wave extends with appreciable amplitude across the rare medium into a nearby region occupied by a higher index material, energy may flow through the gap in what is known as frustrated total internal reflection (FTIR)



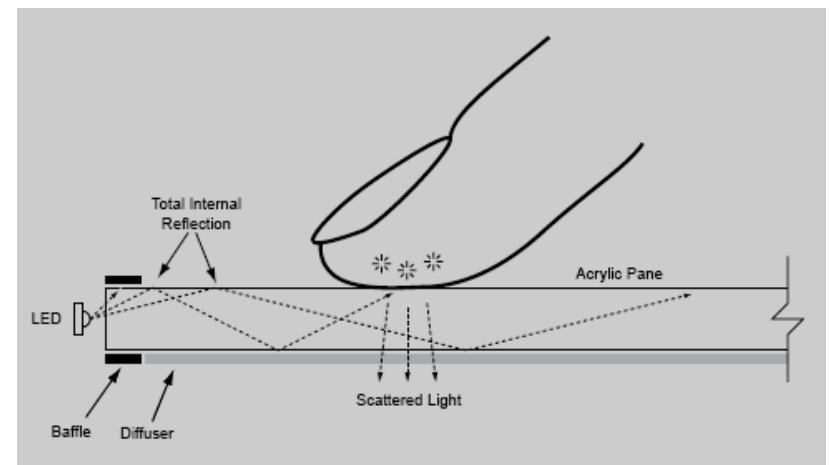
Application: Beamsplitter



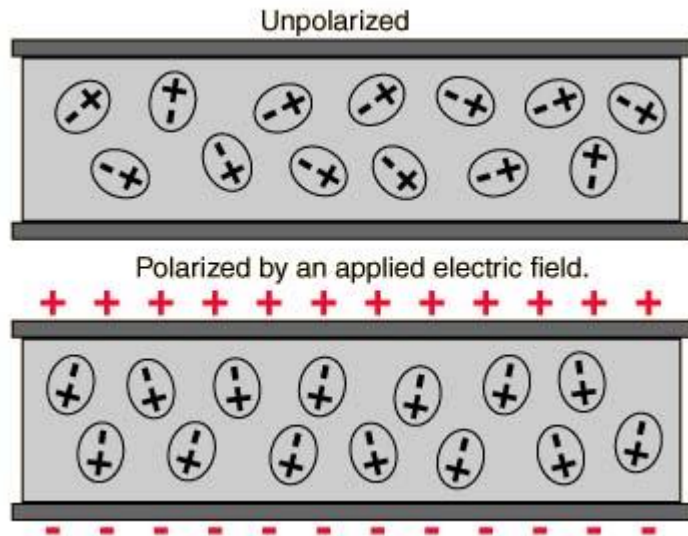
Fingerprint sensors



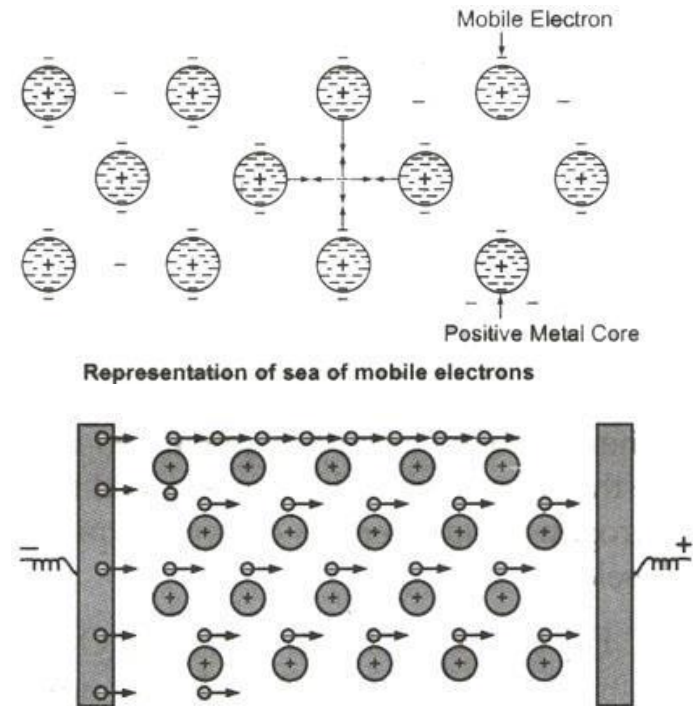
Multi-Touch Sensing through Frustrated Total Internal Reflection



3.8 Optical Properties of Metals



No free or conduction electrons
Conductivity $\sigma = 0$



Electrical conductivity in metal crystals

$$\text{Current density } \vec{j} = \sigma \vec{E}$$

The electrons, driven into oscillation by a harmonic wave, would simply follow the field's alternations. There would be no restoring force, no natural frequencies, and no absorption, only re-emission.

In real metals the conduction electrons undergo collisions with the thermally agitated lattice or with imperfections and in so doing irreversibly convert electromagnetic energy onto joule heat.

Maxwell's equations in conductors

□ Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

□ Gauss's Law- Electric

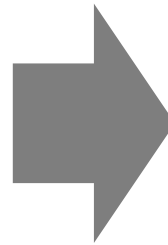
$$\oiint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

□ Gauss's Law- Magnetic

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

□ Ampere-Maxwell's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$



$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon}} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\left\{ \begin{array}{l} \nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{E}) = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} \end{array} \right.$$

$$\nabla^2 \vec{E} = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \boxed{\mu\sigma \frac{\partial \vec{E}}{\partial t}} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

It likes the damping force in the oscillator model

The time rate-of-change of E generates a voltage, current circulate, and since the material is resistive, light is converted to thermal energy-ergo absorption.

The solution for a wave propagating in the y-direction:

$$\vec{E} = \vec{E}_0 \cos \omega(t - \tilde{n}y/c) \quad \tilde{n} = n_R - i n_I$$

$$\vec{E} = \vec{E}_0 e^{-\omega n_I y/c} \cos \omega(t - n_R y/c)$$

The disturbance advances in the y-direction with a speed c/n_R , its amplitude is exponentially attenuated.

Irradiance

Absorption coefficient: $\alpha \equiv 2\omega n_I/c$

$$I(y) = I_0 e^{-\alpha y}$$

Skin depth: $y \equiv 1/\alpha$

Example: copper

For ultraviolet wavelength ($\lambda_0 = 100 \text{ nm}$): Skin depth $y = 0.6 \text{ nm}$

For infrared wavelength ($\lambda_0 = 10000 \text{ nm}$): Skin depth $y = 6 \text{ nm}$

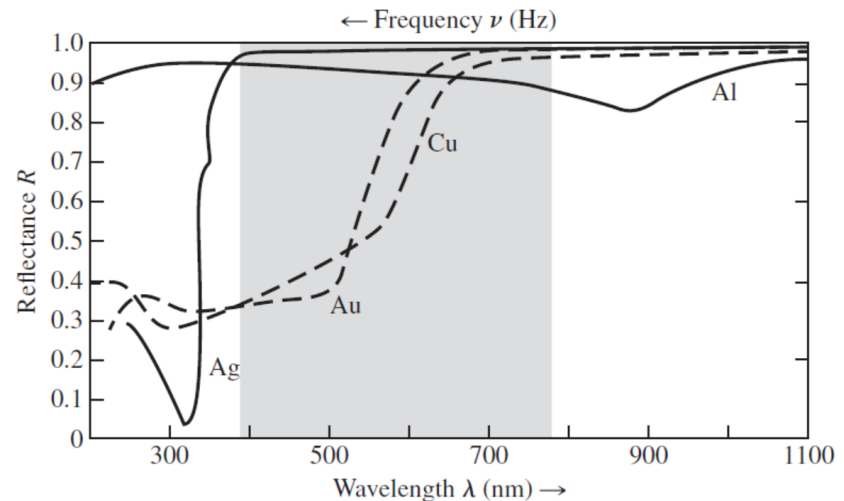
The familiar metallic sheen of conductors corresponds to a high reflectance, which exists because the incident wave cannot effectively penetrate the material.



Relatively few electrons in the metal “see” the transmitted wave, and therefore, although each absorbs strongly, little total energy is dissipated by them. Instead, most of the incoming energy reappears as the reflected wave.

For the simplest case of normal incidence on a metal

$$R = \frac{I_r}{I_i} = \left(\frac{\tilde{n} - 1}{\tilde{n} + 1} \right) \left(\frac{\tilde{n} - 1}{\tilde{n} + 1} \right)^*$$
$$= \frac{(n_R - 1)^2 + n_I^2}{(n_R + 1)^2 + n_I^2}$$



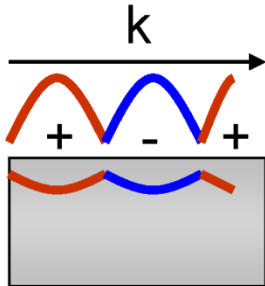
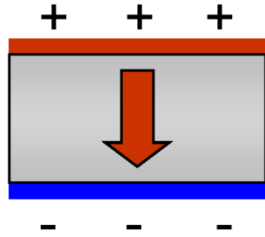
What is Plasmon?

Plasmons = density fluctuation of free electrons

Bulk plasmons

Plasmons in the bulk oscillate at ω_p determined by the free electron density and effective mass

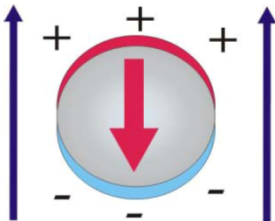
$$\omega_p^{drude} = \sqrt{\frac{Ne^2}{m\epsilon_0}}$$



Surface plasmon polaritons

Plasmons confined to surfaces that can interact with light to form propagating “surface plasmon polaritons (SPP)”

$$k_{sp} = \frac{\omega}{c} \sqrt{\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}}$$



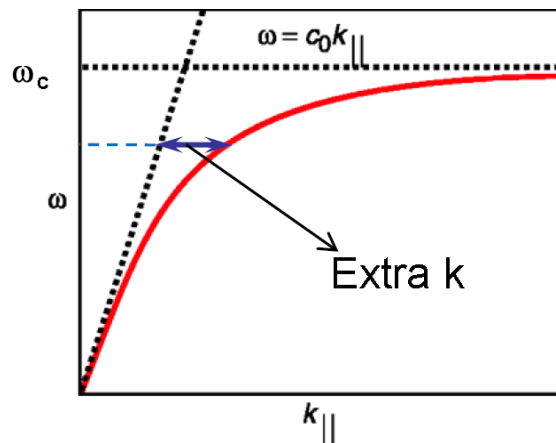
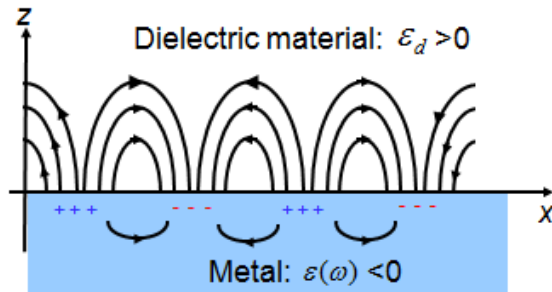
Localized plasmons

Confinement effects result in resonant SPP modes in nanoparticles

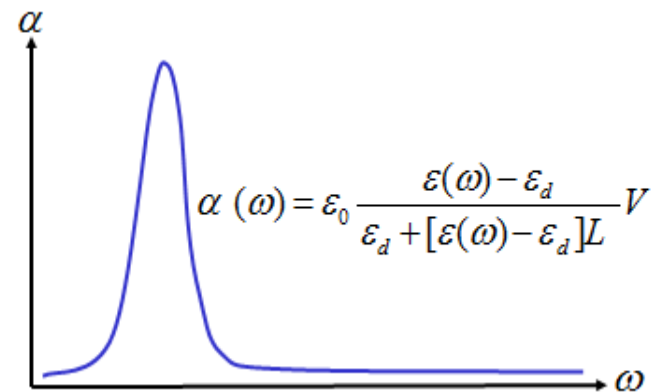
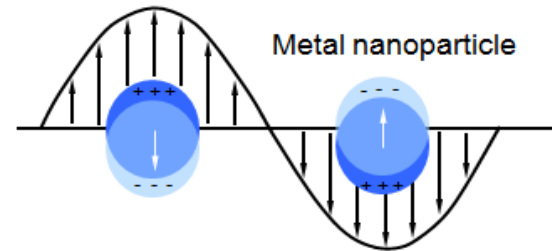
$$\omega_{particle}^{drude} = \sqrt{\frac{1}{3} \frac{Ne^2}{m\epsilon_0}}$$

Plasmonics: Optics at the Nanoscale

Surface plasmon polaritons (SPP)



Localized surface plasmon resonances (LSPR)



❖ Compared to 'regular' photons, SPs have a much smaller wavelength at the same frequency. For example, He-Ne laser: 633 nm, excited SP at a Si/Ag interface: 70 nm.

Overview of Applications of Surface Plasmons

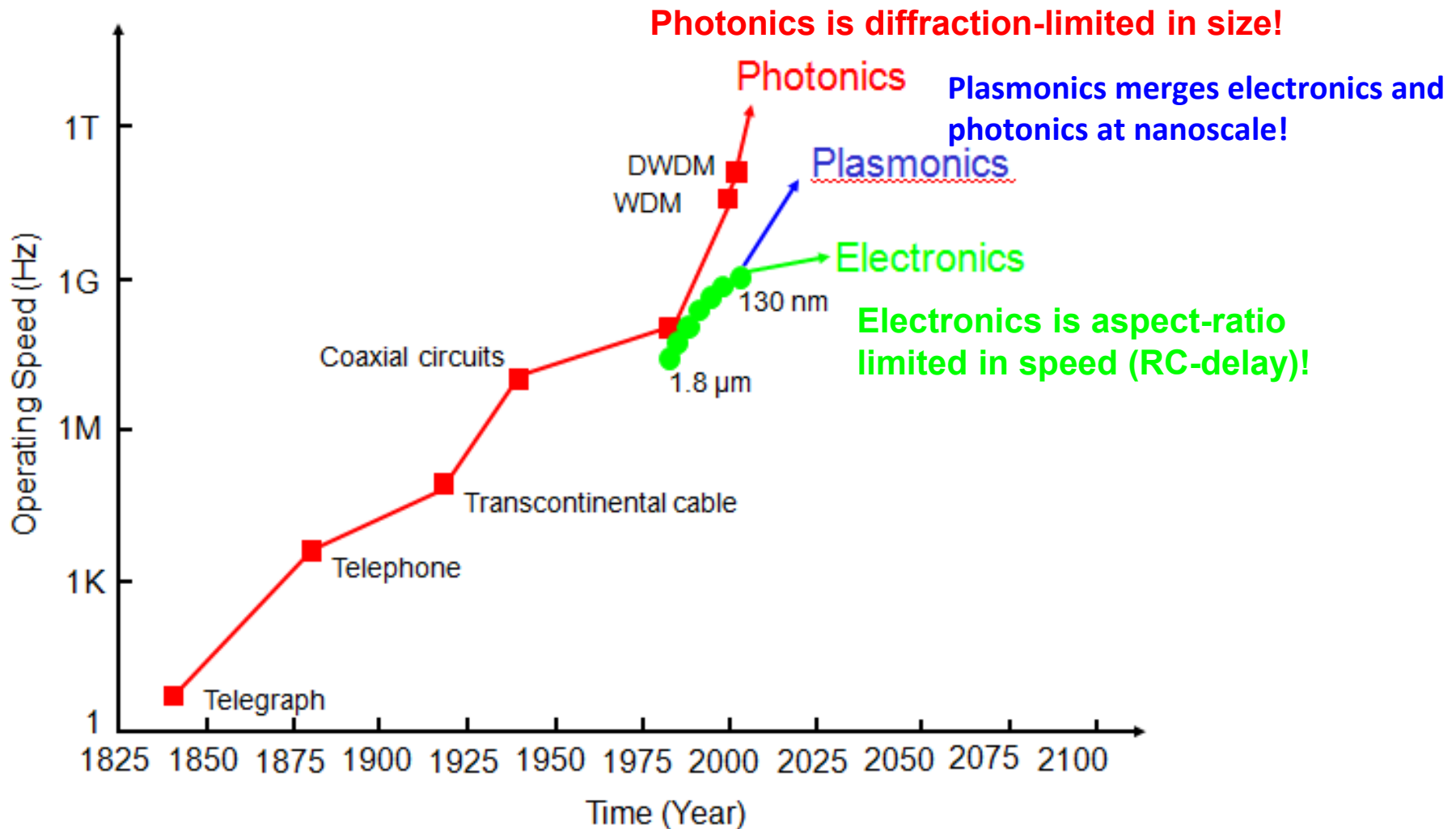
- Optical signal processing
- Metamaterials (extensive light manipulation at subwavelength scales)
- Super-resolution imaging and superfocusing

All above applications are based on

- ✓ Signal amplification (laser) ~~Surface plasmon (SP) energy in deep subwavelength~~
- Light emitting devices ~~Light emitting devices~~
- ✓ Solar cells ~~Solar cells~~ can produce enormous enhancement of the light intensity
- ✓ Optical manipulation ~~Optical manipulation~~ produce enormous gradients of the light intensity
- ✓ Medical applications – drug delivery ~~Medical applications – drug delivery~~
- Metals (gap) ~~Metals (gap)~~
- Medical applications – bio-molecule ultrasensitive sensing
- Medical applications – thermal cell-apoptosis

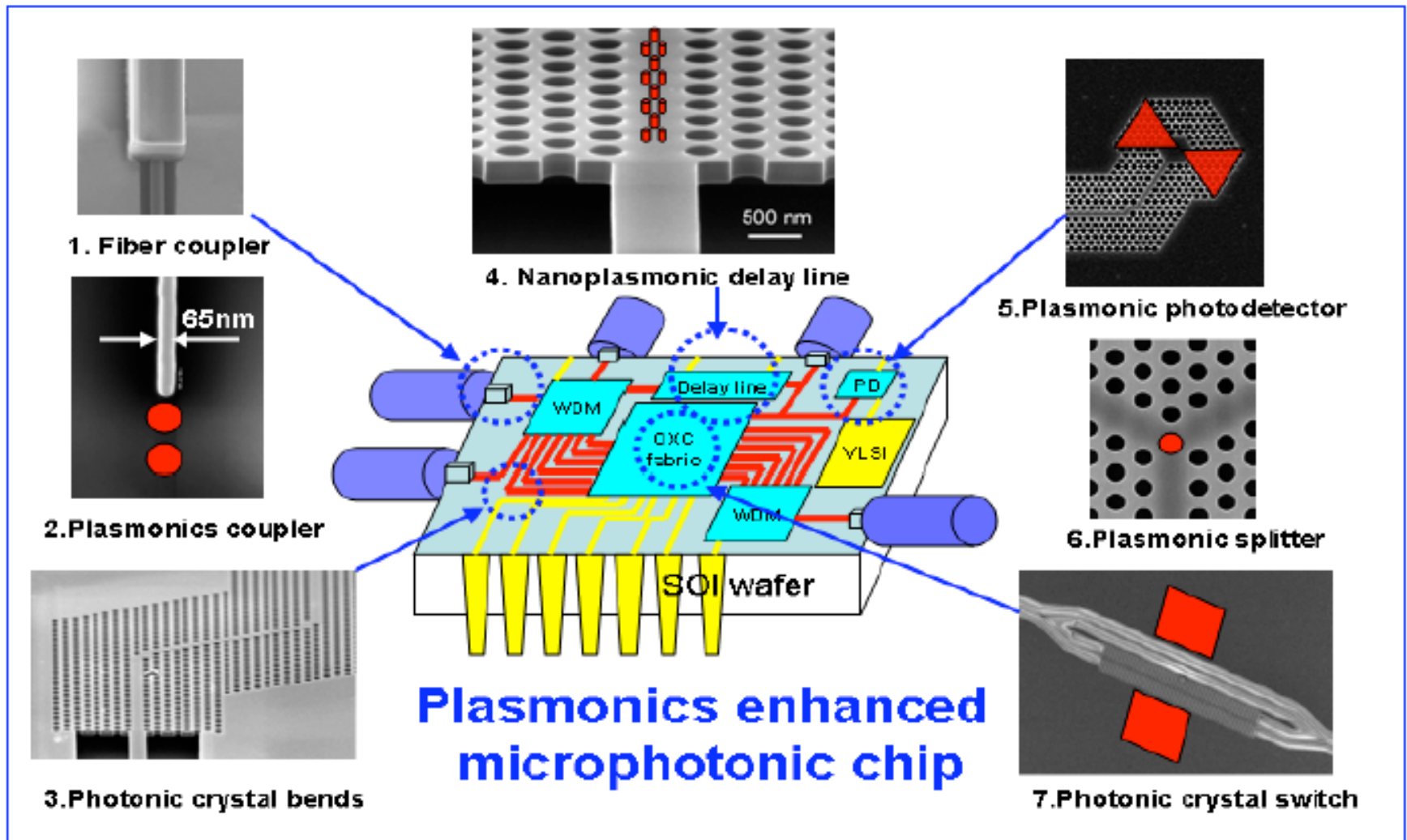
...

Plasmonics is the Next Logic Step in Computing and Information Processing



(Courtesy of Prof. Brongersma@Stanford)

Future Photonic Integrated Chips



Both *passive* and *active* devices are needed.

(Courtesy of Prof. Shalaev@Purdue)

Homework

□ 4.14, 4.39, 4.50, 4.66, 4.70, 4.72, 4.76